Answer Key

9702 Physics · Paper 4 · 2024–2026 · By Topic

答案册（按 topic 分）

12 Motion in a circle

9702/42/M/J/24 Q1 — Disc with modelling clay — angular speed & centripetal acceleration [10 m

(a) [1] The angle subtended at the centre of a circle by an arc whose length is equal to the radius.

(b)(i) [1] Draw an arrow labelled V from the modelling clay pointing in the NE direction — the velocity is tangential to the circular path (perpendicular to the radius of the disc).

(b)(ii) [1] Draw an arrow labelled A from the modelling clay pointing in the NW direction, i.e. towards the centre of the disc (the direction of the centripetal acceleration).

(c)(i) [2] Distance of the clay from the axis: $r = 9.3 - 1.2 = 8.1$ cm $= 8.1\times10^{-2}$ m

$v = r\omega$

$\omega = v/r = 0.68 / 8.1\times10^{-2} = 8.4$ rad s^{-1}

(c)(ii) [2] $a = v^2/r = 0.68^2 / 8.1\times10^{-2} = 0.4624 / 8.1\times10^{-2}$

$= 5.7$ m s^{-2}

(d) [3] angular speed: same for both pieces — the disc is one rigid body, so every point has the same ω.

linear speed: less for the second piece — $v = r\omega$ and the second piece is nearer the axis.

acceleration: less for the second piece — $a = r\omega^2$ and the second piece has the smaller radius.

9702/42/F/M/25 Q1 — Steel ball inside a hollow cone — circular motion & angular speed [9 mark

(a) [1] An arrow vertically downwards labelled ‘weight’, and an arrow perpendicular to the cone surface, pointing inwards and upwards, labelled ‘normal contact force’ (the two arrows must be in the correct directions and correctly labelled).

(b) [2] The vertical component of the contact force equals the weight, so there is no resultant force vertically. The horizontal component of the contact force is the resultant force, and it acts towards the centre of the circle — this provides the centripetal force that keeps the ball moving in a circle.

(c) [3] For a conical-pendulum type of motion the acceleration is $a = g \tan 52°$, and $a = v^2/r$:

$9.81 \times \tan 52° = v^2/0.15$

$v^2 = 0.15 \times 9.81 \times \tan 52° = 1.88$ m^2 s^{-2}

$v = 1.4$ m s^{-1}

(d) [2] $v = r\omega$, so $\omega = v/r = 1.4/0.15$

$= 9.3$ rad s^{-1}

(Equivalently $\omega = \sqrt{(g \tan 52°/0.15)} = 9.3$ rad s^{-1}.)

(e) [1] The resultant (centripetal) force — hence the acceleration — is unchanged, and $a = v^2/r$, so v^2 is proportional to r. Therefore increasing the speed requires a larger radius, so the ball moves up the cone to a circle of larger radius.

9702/42/M/J/25 Q1 — Bicycle chain & cogs — radian, angular speed & period [10 marks]

(a) [1] The angle subtended at the centre of a circle by an arc whose length is equal to the radius.

(b)(i) [2] $v = r\omega$, so $\omega = v/r = 17/0.46$

$= 37$ rad s^{-1}

(b)(ii) [2] $T = 2\pi r/v = 2\pi \times 0.46/17$ (or $T = 2\pi/\omega = 2\pi/37$)

$= 0.17$ s

(b)(iii) [1] distance moved = circumference of the small cog $= 2\pi r = 2\pi \times 0.038$

$= 0.24$ m

(b)(iv) [2] angle = arc length/radius $= 0.24/0.15$

$= 1.6$ rad

(c) [2] The rear wheel still rotates at the same angular speed (the bicycle speed is unchanged), but the rear cog is now smaller, so point X on the chain moves through a smaller distance in the same time — the linear speed of the chain decreases. The large cog (fixed to the pedals) has the same radius, so the angular speed of the pedals decreases.

9702/41/O/N/25 Q1 — Circular motion & simple harmonic motion — shadow projection [15 marks

(a) [2] The velocity and acceleration both have constant magnitude, and the velocity is always perpendicular to the acceleration (the velocity is tangential and the acceleration is directed towards the centre).

(b)(i) [1] $v = R\omega$

(b)(ii) [2] $a = R\omega^2 = (R\omega)\omega = v\omega$

(c)(i) [1] $x = R \sin \theta$

(c)(ii) [1] $\theta = \omega t$

(c)(iii) [1] Substituting $\theta = \omega t$ into $x = R \sin \theta$ gives $x = R \sin \omega t$.

(c)(iv) [1] The equation is of the form $x = x_0 \sin \omega t$, the defining equation of simple harmonic motion (with amplitude $x_0 = R$), so the shadow's motion is simple harmonic.

(d)(i) [1] amplitude = diameter/2 = 0.46/2 = 0.23 m

(d)(ii) [2] $\omega = 2\pi/T$, so $T = 2\pi/\omega = 2\pi/1.9$

$= 3.3$ s

(d)(iii) [2] $a_0 = \omega^2 x_0 = 1.9^2 \times 0.23$

$= 0.83$ m s^{-2}

(e) [1] The shadow is at the extreme of its motion — at the left-hand edge of the circular path (maximum displacement from P), labelled A, where the acceleration is a maximum.

9702/42/O/N/25 Q1 — Circular motion of a student at latitude 52.2° [9 marks]

(a)(i) [1] radius $= 6.37 \times 10^6 \times \cos 52.2°$

$= 3.90 \times 10^6$ m

(a)(ii) [3] $v = 2\pi r/T = 2\pi \times 3.90 \times 10^6/(24 \times 60 \times 60)$

$= 280$ m s^{-1}

(b)(i) [2] $F = mv^2/r = 58.6 \times 280^2/(3.90 \times 10^6)$

$= 1.2$ N

(b)(ii) [1] An arrow pointing horizontally to the left (towards the Earth's axis of rotation).

(b)(iii) [2] An arrow downwards from the student (towards the centre of the Earth) labelled 'weight', and an upwards arrow (the normal contact force) pointing slightly to the left of the normal, labelled 'contact force' — their resultant is the horizontal centripetal force.

9702/43/O/N/25 Q1 — Circular motion & simple harmonic motion — shadow projection [15 marks

(a) [2] The velocity and acceleration both have constant magnitude, and the velocity is always perpendicular to the acceleration (the velocity is tangential and the acceleration is directed towards the centre).

(b)(i) [1] $v = R\omega$

(b)(ii) [2] $a = R\omega^2 = (R\omega)\omega = v\omega$

(c)(i) [1] $x = R \sin \theta$

(c)(ii) [1] $\theta = \omega t$

(c)(iii) [1] Substituting $\theta = \omega t$ into $x = R \sin \theta$ gives $x = R \sin \omega t$.

(c)(iv) [1] The equation is of the form $x = x_0 \sin \omega t$, the defining equation of simple harmonic motion (with amplitude $x_0 = R$), so the shadow's motion is simple harmonic.

(d)(i) [1] amplitude = diameter/2 = 0.46/2 = 0.23 m

(d)(ii) [2] $\omega = 2\pi/T$, so $T = 2\pi/\omega = 2\pi/1.9$

$= 3.3$ s

(d)(iii) [2] $a_0 = \omega^2 x_0 = 1.9^2 \times 0.23$

$= 0.83$ m s^{-2}

(e) [1] The shadow is at the extreme of its motion — at the left-hand edge of the circular path (maximum displacement from P), labelled A, where the acceleration is a maximum.

9702/42/M/J/26 Q1 — Aircraft in horizontal circle — centripetal acceleration & banking angle [10 marks]

(a) [2] The acceleration is (always) perpendicular to the velocity, directed towards the centre of the circle, and the magnitudes of the velocity (the speed) and of the acceleration are constant — but their directions change continuously.

(b)(i) [2] The only other force on the aircraft is its weight, which is vertical and has no horizontal component. The lift must therefore have a horizontal component (the resultant force towards the centre of the circle) to provide the centripetal force — so the lift cannot be vertically upwards.

(b)(ii) [1] Arrow drawn from the aircraft pointing horizontally to the left, i.e. towards the centre of the circle — the resultant force provides the centripetal force.

(b)(iii) [2] $a = v^2/r = 180^2/7200$

$= 4.5$ m s^{-2}

(b)(iv) [3] The vertical component of the lift balances the weight ($L \cos \theta = mg$) and the horizontal component of the lift causes the centripetal acceleration ($L \sin \theta = ma$):

$\tan \theta = a/g = 4.5/9.81$

$\theta = 25°$

13 Gravitational fields

9702/42/F/M/24 Q1 — Gravitational potential of a planet — satellite orbit & escape energy [10 marks]

(a) [2] Gravitational potential is defined to be zero at infinity. The gravitational force is attractive, so work must be done on a mass to move it from a point in the field to infinity — the potential therefore rises towards zero from negative values, so near a point mass it is negative.

(b)(i) [2] $\varphi = -GM/r$, so $M = -\varphi r/G$. From Fig. 1.1, at $1/r = 3.75\times10^{-8}$ m^{-1} the potential is $\varphi = -2.2\times10^{8}$ J kg^{-1}:

$M = (2.2\times10^{8}/3.75\times10^{-8})/6.67\times10^{-11} = 8.8\times10^{25}$ kg

(b)(ii) [3] $GMm/R^2 = mR\omega^2$ with $\omega = 2\pi/T \rightarrow R^3 = GMT^2/4\pi^2$

$T = 0.72\times24\times3600 = 6.22\times10^{4}$ s

$R^3 = 6.67\times10^{-11}\times8.8\times10^{25}\times(6.22\times10^{4})^2/4\pi^2 = 5.75\times10^{23}$

$R = 8.3\times10^{7}$ m

(b)(iii) [3] Energy required = ΔGPE − KE = $GMm/R - \frac{1}{2}mv^2$

$= 6.67\times10^{-11}\times8.8\times10^{25}\times1200/(8.3\times10^{7}) - \frac{1}{2}\times1200\times8400^2$

$= 8.49\times10^{10} - 4.23\times10^{10} = 4.3\times10^{10}$ J

9702/41/M/J/24 Q1 — Satellite in circular orbit — gravitational field strength & energy [10 marks

(a) [2] The work done per unit mass
in bringing a small test mass from infinity to the point.

(b)(i) [2] The gravitational potential is defined as zero at infinity. As the two masses move closer together (or as a mass is brought towards the planet), work is done by the gravitational field, so the potential falls below zero — hence it is negative.

(b)(ii) [2] $\phi = -GM/r$ and, at X, $-\Phi = -GM/(4R)$, so $GM/R = 4\Phi$.

At Y (radius R): potential = -4Φ

(b)(iii) [4] Since $\Phi = GM/(4R)$, $GM = 4\Phi R$.

Field strength: $g = GM/r^2$

X: $g = GM/(4R)^2 = \Phi/4R$; Y: $g = GM/R^2 = 4\Phi/R$

Potential energy: $EP = -GMm/r$

X: $EP = -GM\cdot M/4R = -M\Phi$; Y: $EP = -GM\cdot2M/R = -8M\Phi$

9702/41/O/N/24 Q1 — Satellite in circular orbit — Newton's law & orbital period graph [12 marks

(a) [2] The gravitational force between two point masses is directly proportional to the product of the masses and inversely proportional to the square of the distance between them: $F = Gm_1m_2/r^2$.

(b)(i) [2] The gravitational force on the satellite acts towards the centre of the planet, so it is always perpendicular to the direction of motion of the satellite. This force provides the centripetal acceleration, so the satellite moves in a circular path at constant speed.

(b)(ii) [3] Gravitational force = centripetal force:

$GMm/(h + B)^2 = m(h + B)\omega^2$, with $\omega = 2\pi/T$

$GM/(h + B)^2 = (h + B)(2\pi/T)^2$

$(h + B)^3 = GMT^2/4\pi^2 = (GA/4\pi^2)T^2$

where A = M (the mass of the planet) and B = the radius of the planet.

(b)(iii) [5] From Fig. 1.1, gradient $m = (1280 - 360)/(12\times10^{6}) = 7.7\times10^{-5}$ s$^{2/3}$ m^{-1}.

Since $T^{2/3} = m(h + B)$, the intercept is mB, and $m^3 = 4\pi^2/GA$.

$A = 4\pi^2/(Gm^3) = 4\pi^2/(6.67\times10^{-11}\times(7.7\times10^{-5})^3) = 1.3\times10^{24}$ kg

B = intercept/m = $360/(7.7\times10^{-5}) = 4.7\times10^{6}$ m

9702/42/O/N/24 Q2 — Probe approaching the Sun — field strength, potential & luminosity [12 m

(a) [1] Gravitational field (strength) is the force per unit mass acting on a small test mass placed in the field.

(b)(i) [2] $g = GM/x^2 = (6.67\times10^{-11}\times1.99\times10^{30})/(1.47\times10^{11})^2$

$= 6.14\times10^{-3}$ N kg^{-1}

(b)(ii) [2] $EP = -GMm/x = -(6.67\times10^{-11}\times1.99\times10^{30}\times2.63)/(1.47\times10^{11})$

$= -2.37\times10^{9}$ J

(c)(i) [3] $F = L/(4\pi x^2)$, so $x^2 = L/(4\pi F)$, and $g = GM/x^2$, so $x^2 = GM/g$.

Eliminating x^2: $GM/g = L/(4\pi F)$, giving $g = 4\pi GMF/L$.

(c)(ii) [2] Read from Fig. 2.1 a consistent pair of values, e.g. $F = 1.83\times10^{3}$ W m^{-2} when $g = 8.0\times10^{-3}$ N kg^{-1}.

$L = 4\pi GMF/g = (4\pi \times 6.67\times10^{-11} \times 1.99\times10^{30} \times 1.83\times10^{3})/(8.0\times10^{-3})$

$= 3.8\times10^{26}$ W

(c)(iii) [2] $L = 4\pi\sigma r^2T^4 \rightarrow r = \sqrt{(L/(4\pi\sigma T^4))}$

$= \sqrt{(3.8\times10^{26}/(4\pi \times 5.67\times10^{-8} \times 5780^4))} = \sqrt{(4.78\times10^{17})}$

$= 6.9\times10^{8}$ m

9702/42/F/M/25 Q2 — Gravitational potential of a planet & electric potential between plates [7 marks]

(a) [3] Sketch φ against x: a curve lying entirely in the negative-φ region for x between R and 4R, starting at (R, −φ) with a continuously decreasing magnitude and continuously decreasing gradient (concave up towards zero), and passing through (2R, −½φ) and (4R, −¼φ). The potential is $\varphi = -GM/x$, so it halves as x doubles.

(b) [2] A geostationary satellite stays at the same point above the Earth, so its gravitational potential is constant: a horizontal straight line from t = 0 to t = 24 hours, starting at (0, −φ).

(c) [2] The field between the plates is uniform, so the potential falls uniformly: a straight line with constant negative gradient from (0, V) to (d, 0), i.e. the potential decreases linearly from the positive plate to the negative plate.

9702/41/M/J/25 Q1 — Gravitational potential & field of Mars — satellite orbit [9 marks]

(a) [2] The work done per unit mass
in moving a small test mass from infinity to the point.

(b)(i) [3] The satellite moves from $r_1 = (3.4 + 1.7) \times 10^6 = 5.1 \times 10^6$ m to $r_2 = (3.4 + 6.8) \times 10^6 = 10.2 \times 10^6$ m from the centre of Mars. Using $\Delta EP = GMm(1/r_1 - 1/r_2)$:

$5.1 \times 10^8 = 6.67 \times 10^{-11} \times M \times 122 \times [(5.1 \times 10^6)^{-1} - (10.2 \times 10^6)^{-1}]$

$M = 6.4 \times 10^{23}$ kg

(b)(ii) [2] $\varphi = -GM/R = -(6.67 \times 10^{-11} \times 6.4 \times 10^{23})/(3.4 \times 10^6)$

$= -1.3 \times 10^7$ J kg^{-1}

(c)(i) [1] The satellite stays over the same point of Mars, so its period equals Mars's rotation period: Mars takes (just under) 25 hours to rotate once on its axis.

(c)(ii) [1] The orbit is equatorial, and/or the satellite orbits in the same direction as the rotation of Mars.

9702/43/M/J/25 Q1 — Gravitational potential & field of Mars — satellite orbit [9 marks]

(a) [2] The work done per unit mass
in moving a small test mass from infinity to the point.

(b)(i) [3] The satellite moves from $r_1 = (3.4 + 1.7) \times 10^6 = 5.1 \times 10^6$ m to $r_2 = (3.4 + 6.8) \times 10^6 = 10.2 \times 10^6$ m from the centre of Mars. Using $\Delta EP = GMm(1/r_1 - 1/r_2)$:

$5.1 \times 10^8 = 6.67 \times 10^{-11} \times M \times 122 \times [(5.1 \times 10^6)^{-1} - (10.2 \times 10^6)^{-1}]$

$M = 6.4 \times 10^{23}$ kg

(b)(ii) [2] $\varphi = -GM/R = -(6.67 \times 10^{-11} \times 6.4 \times 10^{23})/(3.4 \times 10^6)$

$= -1.3 \times 10^7$ J kg^{-1}

(c)(i) [1] The satellite stays over the same point of Mars, so its period equals Mars's rotation period: Mars takes (just under) 25 hours to rotate once on its axis.

(c)(ii) [1] The orbit is equatorial, and/or the satellite orbits in the same direction as the rotation of Mars.

9702/44/M/J/25 Q1 — Gravitational field & potential of the Earth [11 marks]

(a) [1] Gravitational field is the force per unit mass acting on a small test mass placed at the point.

(b)(i) [2] The field lines are radial, pointing towards the centre of the planet.

(b)(ii) [2] For small changes in height, the change in height is very much smaller than the radius of the planet, so $(R + h)^2 \approx R^2$ — the field lines are approximately parallel and g is approximately constant.

(c)(i) [2] At the surface $g = GM/R^2$, so $9.81 = 3.99 \times 10^{14}/R^2$

$R = \sqrt{(3.99 \times 10^{14}/9.81)} = 6.38 \times 10^{6}$ m

(c)(ii) [2] $\phi = -GM/R = -3.99 \times 10^{14}/(6.38 \times 10^{6})$
$= -6.25 \times 10^{7}$ J kg^{-1}

(d) [2] The potential energy is zero at infinite separation, and the gravitational force is attractive — work is done by the field as the masses come together, so the potential energy falls below zero and is always negative.

9702/41/O/N/25 Q3 — Gravitational field — point masses & two spheres [9 marks]

(a) [1] The force per unit mass acting on a small test mass placed at the point.

(b)(i) [2] $F = GMm/x^2$ and $g = F/m$, so $g = [GMm/x^2]/m = GM/x^2$, where G is the gravitational constant.

(b)(ii) [1] An arrow at P pointing directly towards the point mass.

(b)(iii) [2] The field at Q is in the opposite direction to that at P (towards the mass), and is four times the field strength at P (Q is at distance x/2, and $g \propto 1/x^2$).

(c) [3] A line starting at $(R, -g_0)$ and ending at $(L - R, +g_0)$, passing through (L/2, 0), becoming shallower from R to L/2 and steeper again from L/2 to L − R.

9702/43/O/N/25 Q3 — Gravitational field — point masses & two spheres [9 marks]

(a) [1] The force per unit mass acting on a small test mass placed at the point.

(b)(i) [2] $F = GMm/x^2$ and $g = F/m$, so $g = [GMm/x^2]/m = GM/x^2$, where G is the gravitational constant.

(b)(ii) [1] An arrow at P pointing directly towards the point mass.

(b)(iii) [2] The field at Q is in the opposite direction to that at P (towards the mass), and is four times the field strength at P (Q is at distance x/2, and $g \propto 1/x^2$).

(c) [3] A line starting at $(R, -g_0)$ and ending at $(L - R, +g_0)$, passing through (L/2, 0), becoming shallower from R to L/2 and steeper again from L/2 to L − R.

9702/44/O/N/25 Q1 — Binary star — Newton's law, orbit & centripetal acceleration [11 marks]

(a) [2] The gravitational force is (directly) proportional to the product of the two masses and inversely proportional to the square of their separation.

(b)(i) [2] $F = Gm_1m_2/r^2 = 6.67 \times 10^{-11} \times 4.0 \times 10^{30} \times 2.0 \times 10^{30}/(3.3 \times 10^{12})^2$
$= 4.9 \times 10^{25}$ N

(b)(ii) [1] $a = F/m = 4.9 \times 10^{25}/4.0 \times 10^{30}$
$= 1.2 \times 10^{-5}$ m s^{-2}

(b)(iii) [3] $a = r\omega^2$ with $\omega = 2\pi/T$. Since RB = 2RA and RA + RB = 3.3×10^{12} m, RA = $3.3 \times 10^{12}/3 = 1.1 \times 1$ m.

$T = 2\pi\sqrt{(r/a)} = 2\pi\sqrt{(1.1 \times 10^{12}/1.2 \times 10^{-5})}$
$= 1.9 \times 10^{9}$ s

(b)(iv) [3] centripetal acceleration: B greater than A (a = F/m, and B has the smaller mass); linear speed: B greater than A (v = rω, and B has the larger radius); period: B equal to A (both orbit X with the same period).

9702/41/M/J/26 Q1 — Gravitational potential & field strength of a uniform planet [10 marks]

(a) [2] The work done per unit mass
in bringing a small test mass from infinity to the point.

(b)(i) [2] The gravitational potential is defined as zero at infinity. Work is done by the two masses as they move towards each other (equivalently, work must be done on them to move them apart), so the potential decreases below zero — hence φ is negative for all values of x.

(b)(ii) [2] $\phi = -GM/r$. At the surface, r = R and $\phi = -\Phi$, so $-\Phi = -GM/R$.
$M = \Phi R/G$, where G is the (universal) gravitational constant.

(b)(iii) [1] $g_0 = GM/R^2 = (\Phi R)/R^2 = \Phi/R$

(b)(iv) [3] Two curves: one in the top-left quadrant (between x = −3R and x = −R) and one in the bottom-right quadrant (between x = R and x = 3R). The upper curve passes through $(-R, g_0)$ and $(-2R, \frac{1}{4}g_0)$, with a non-zero value of g at x = −3R; the lower curve passes through $(R, -g_0)$ and $(2R, -\frac{1}{4}g_0)$, with a non-zero value of g at x = 3R. Between −3R and −R the curve has a gradient and a magnitude of g that increase towards −R; between R and 3R the gradient and the magnitu of g decrease away from R. Nothing is drawn between x = −R and x = R.

9702/42/M/J/26 Q2 — Earth's gravitational field — mass, potential & satellite energy [9 marks]

(a)(i) [1] The gravitational field (strength) at a point is the force per unit mass acting on a small test

mass placed at that point.

(a)(ii) [2] Draw four straight, parallel field lines perpendicular to the ground (vertical), approximately evenly spaced, with arrows pointing downwards towards the ground.

(b)(i) [2] $g = GM/R^2 \rightarrow M = gR^2/G = 9.81\times(6.37\times10^6)^2/(6.67\times10^{-11})$
$= 5.97\times10^{24}$ kg

(b)(ii) [2] $\varphi = -GM/R = -6.67\times10^{-11}\times5.97\times10^{24}/(6.37\times10^6)$
$= -6.25\times10^7$ J kg^{-1}

(b)(iii) [2] Orbital radius $r = (6.37 + 1.93)\times10^6 = 8.30\times10^6$ m
$E_P = -GMm/r = -6.67\times10^{-11}\times5.97\times10^{24}\times96.0/(8.30\times10^6)$
$= -4.61\times10^9$ J

9702/43/M/J/26 Q1 — Gravitational potential & field strength of a uniform planet [10 marks]

(a) [2] The gravitational potential at a point is the work done per unit mass in bringing a small test mass from infinity to that point.

(b)(i) [2] The potential is defined as zero at infinity.
Work is done by the two masses as they move towards each other from infinity (or work must be done on them to increase their separation), so their gravitational potential energy decreases — hence the potential at any finite point is negative for all values of x.

(b)(ii) [2] At the surface: $\phi = -GM/R = -\Phi$
$M = \Phi R/G$, where G is the universal gravitational constant.

(b)(iii) [1] $g0 = GM/R^2 = G(\Phi R/G)/R^2 = \Phi/R$

(b)(iv) [3] Draw two curves — one in the top-left quadrant and one in the bottom-right quadrant (nothing drawn between x = −R and x = +R).
Each curve must pass through (±R, ±g0) and (±2R, ±¼g0), with a non-zero value of g at ±3R.
The magnitudes of both g and the gradient decrease as |x| increases ($g \propto 1/x^2$), so each curve flattens towards the x-axis away from the planet.

9702/44/M/J/26 Q2 — Geostationary satellite — Earth's mass & mean density [10 marks]

(a)(i) [1] A geostationary orbit is one in which the satellite remains above the same point on the Earth's surface.

(a)(ii) [2] Any two of: the orbit is equatorial (directly above the Equator); the satellite orbits from west to east; the orbital period is 24 hours (equal to the Earth's period of rotation).

(b)(i) [1] $v = 2\pi R/T$

(b)(ii) [2] $GMm/R^2 = mR\omega^2$ and $\omega = 2\pi/T$:
$GMm/R^2 = mR(2\pi/T)^2 \rightarrow M = 4\pi^2R^3/(GT^2)$

(c) [4] Orbit radius $R = (6370 + 35\,900)\times10^3 = 4.23\times10^7$ m; $T = 24\times3600 = 8.64\times10^4$ s
$M = 4\pi^2R^3/(GT^2) = 4\pi^2\times(4.23\times10^7)^3/(6.67\times10^{-11}\times(8.64\times10^4)^2) = 5.99\times10^{24}$ kg
$V = (4/3)\pi r^3 = (4/3)\pi\times(6.37\times10^6)^3 = 1.08\times10^{21}$ m^3
density $= M/V = 5.99\times10^{24}/1.08\times10^{21} = 5530$ kg m^{-3}

14 Temperature

9702/41/M/J/24 Q2 — Temperature measurement — absolute zero, platinum thermometer & ther marks]

(a)(i) [1] Magnitude 0 K (zero kelvin), the lowest possible temperature on the thermodynamic scale.

(a)(ii) [1] Its measurement depends on the properties of the liquid (its expansion), whereas the thermodynamic scale does not depend on the properties of any particular substance.

(b)(i) [2] Any two of: the resistivity (and hence resistance) of platinum varies with temperature; the variation with temperature is linear; there is a unique value of resistivity for each (different value of) temperature.

(b)(ii) [1] The thermometer has a high heat capacity, so it takes time (energy transfer) to reach the temperature of the surroundings and cannot follow a rapidly changing temperature.

(b)(iii) [1] A thermocouple.

(c) [1] The resistance of the (negative temperature coefficient) thermistor decreases as temperature increases (inverse, non-linear variation), whereas the resistance of a platinum wire increases with temperature.

9702/41/O/N/24 Q2 — Two metal blocks in thermal contact — specific heat capacity [8 marks]

(a) [2] The (thermal) energy required per unit mass of a substance to change its temperature by one unit (1 K): $c = \Delta E/(m\Delta\theta)$.

(b)(i) [3] Any three of (up to two may be quantitative):

1. the two blocks end up in thermal equilibrium — both reach the same final temperature;
2. the heat capacity of Y is larger than that of X (Y's temperature changes less for the same energy transfer);
3. no heat is lost to the surroundings (the energy lost by X equals the energy gained by Y);
4. the initial temperature of X is 85 °C and of Y is 25 °C;
5. the temperature change of X is 45 °C and of Y is 15 °C (the change for X is three times that for Y);
6. the final temperature of both blocks is 40 °C.

(b)(ii) [3] Heat lost by X = heat gained by Y:

$mX cX \Delta\theta X = mY cY \Delta\theta Y$, with $\Delta\theta X = 45$ °C, $\Delta\theta Y = 15$ °C and $mY = 1.3mX$

$mX cX \times 45 = 1.3mX \times 901 \times 15$

$cX = (1.3 \times 901 \times 15)/45 = 390$ J kg^{-1} K^{-1}

9702/42/O/N/24 Q3 — Vaporising liquid — specific latent heat & internal energy [11 marks]

(a) [2] Specific latent heat is the (thermal) energy per unit mass required to cause a change of state, at constant temperature.

(b)(i) [2] $W = p\Delta V = 1.0\times10^5 \times 0.017 = 1700$ J = 1.7 kJ (work done by the substance as it expands, so the work done on the substance is −1.7 kJ, of magnitude 1.7 kJ)

(b)(ii) [2] $\Delta U = Q + W$, where W is the work done on the substance (= −1.7 kJ):

$Q = \Delta U - W = 17.6 - (-1.7)$

$= 19.3$ kJ

(b)(iii) [2] $\text{mass} = \rho V = 710 \times 7.2\times10^{-5} = 0.051$ kg

$LV = Q/m = 19.3/0.051$

$= 380$ kJ kg^{-1}

(c) [3] LF is likely to be less than LV. Fusion involves a much smaller volume change than vaporisation, so the change in intermolecular spacing (and hence the change in internal energy) is much smaller, and the work done by the substance during fusion is negligible — both contributions are far smaller for fusion than for vaporisation.

9702/41/M/J/25 Q3 — Specific latent heat — vaporisation vs fusion & ice–water mixture [9 marks

(a) [2] The (thermal) energy per unit mass required to change state at constant temperature.

(b) [3] Vaporisation involves a much larger increase in the separation (or volume) of the molecules than fusion, so much more work has to be done against the intermolecular forces — a much larger increase in molecular potential energy. Since the kinetic energy of the molecules is unchanged in either change of state, more thermal energy is needed to vaporise a substance.

(c) [4] Heat lost by the water = heat to melt the ice + heat to warm the melted water. Using $Q = mc\Delta\theta$

and Q = mL:

$(37.0 \times L) + (37.0 \times 4.18 \times 10.3) = 208 \times 4.18 \times (26.4 - 10.3)$

$37.0L + 1593 = 14000$

$L = 335\ J\ g^{-1}$

9702/42/M/J/25 Q3 — Specific heat capacity & work done — heated aluminium block [13 marks]

(a) [2] The (thermal) energy per unit mass required to raise the temperature of the substance by unit temperature (1 K or 1 °C).

(b)(i) [2] density = mass/volume, so mass = $\rho V = 2.700 \times 10^3 \times 3.612 \times 10^{-3}$

$= 9.75$ kg

(b)(ii) [1] The mass is constant, so $V = V_0 \times (\rho_0/\rho) = 3.612 \times 10^{-3} \times (2.700/2.620)$

$= 3.722 \times 10^{-3}\ m^3$

(b)(iii) [2] $W = p\Delta V = 1.01 \times 10^5 \times (3.722 - 3.612) \times 10^{-3}$

$= 11.1$ J

(b)(iv) [2] The volume of the block increases, so it pushes against the atmosphere and work is done against the atmosphere — the work done on the block is therefore negative.

(b)(v) [3] From the first law $\Delta U = q + W$, the thermal energy supplied is $q = \Delta U - W = 4.38 \times 10^6 + 11.1$ J (since W is negative).

Specific heat capacity $= q/(m\Delta\theta) = 4.38 \times 10^6/(9.75 \times 500)$

$= 898\ J\ kg^{-1}\ °C^{-1}$

(c) [1] Doubling the pressure would double the work done, but the work done (11.1 J) is negligible compared with the change in internal energy (4.38 MJ), so the specific heat capacity would be unchanged.

9702/43/M/J/25 Q3 — Specific latent heat — vaporisation vs fusion & ice–water mixture [9 marks]

(a) [2] The (thermal) energy per unit mass required to change state at constant temperature.

(b) [3] Vaporisation involves a much larger increase in the separation (or volume) of the molecules than fusion, so much more work has to be done against the intermolecular forces — a much larger increase in molecular potential energy. Since the kinetic energy of the molecules is unchanged in either change of state, more thermal energy is needed to vaporise a substance.

(c) [4] Heat lost by the water = heat to melt the ice + heat to warm the melted water. Using $Q = mc\Delta\theta$ and $Q = mL$:

$(37.0 \times L) + (37.0 \times 4.18 \times 10.3) = 208 \times 4.18 \times (26.4 - 10.3)$

$37.0L + 1593 = 14000$

$L = 335\ J\ g^{-1}$

9702/44/M/J/25 Q2 — Thermal equilibrium — ice melting in water [7 marks]

(a) [2] The two objects are at the same temperature, so there is no net transfer of thermal energy between them.

(b)(i) [1] $E_1 = XL$ (the energy to melt the ice at 0 °C).

(b)(ii) [1] $E_2 = Mc(t - \theta)$ (the energy lost by the water cooling from t to θ).

(b)(iii) [1] $E_3 = Xc\theta$ (the energy gained by the melted ice warming from 0 °C to θ).

(c) [2] Energy is conserved: $E_2 = E_1 + E_3$, so $Mc(t - \theta) = XL + Xc\theta$.

$Mct - Mc\theta = XL + Xc\theta$

$Mct - XL = \theta(Mc + Xc) = \theta c(M + X)$

$\theta = (Mct - XL)/c(M + X)$

9702/42/O/N/25 Q3 — Constant-volume gas thermometer — thermodynamic temperature [9 mar

(a) [2] Two objects are in thermal equilibrium when they are at the same temperature, so there is no net transfer of thermal energy between them.

(b)(i) [1] The density of the liquid (used in $p = \rho g\Delta h$).

(b)(ii) [2] Any two of: a large response time (cannot measure rapidly changing temperatures); reaching equilibrium requires a significant transfer of energy (so it changes the temperature of the environment being measured); it is bulky/difficult to set up and to take readings from.

(b)(iii) [1] To measure the temperature of a substance with a large mass, or a constant temperature, or to calibrate other thermometers.

(b)(iv) [3] 0 °C = 273 K. The temperature is proportional to the pressure (hence Δh):

$T = 273 \times (7.83 - 2.31)/(8.69 - 2.31) = 236$ K

θ = 236 − 273 = −37 °C

9702/44/M/J/26 Q3 — Thermistor in warm water — thermal equilibrium & specific heat capacity [7 marks]

(a) [2] Two objects are in thermal equilibrium when they are at the same temperature, so that (if in thermal contact) there is no net transfer of thermal energy between them.

(b)(i) [2] The thermistor absorbs thermal energy to raise its own temperature; this energy is transferred from the water, so the water cools. The equilibrium temperature is therefore below 78.0 °C.

(b)(ii) [3] Energy lost by water = energy gained by thermistor: $Q = mc\Delta\theta$

$63 \times 4.2 \times (78.0 - 76.8) = 8.5 \times c \times (76.8 - 25.0)$

$317.5 = 440.3c$

$c = 0.72\ \text{J g}^{-1}\ \text{K}^{-1}$

(c) [2] Use a thermocouple, or a smaller/lower-mass thermistor: the smaller mass (heat capacity) removes less thermal energy from the water, so the reading is closer to the true 78.0 °C. Alternatively, warm the thermistor to about 78 °C before immersion, so it removes much less thermal energy from the water.

15 Ideal gases

9702/41/M/J/24 Q3 — Ideal gas — kinetic theory, internal energy & U–V graph [12 marks]

(a)(i) [2] A gas for which pV ∝ T, where T is the thermodynamic temperature (i.e. an ideal gas obeys the equation of state at all temperatures and pressures).

(a)(ii) [2] One basic assumption is that there are no intermolecular forces between the molecules (except during collisions). Hence the potential energy associated with the random motion of the molecules is zero.

(b)(i) [2] pV = NkT

$N = (2.0\times10^{5} \times 0.26) / (1.38\times10^{-23} \times 290) = 5.2\times10^{4} / 4.0\times10^{-21} = 1.3\times10^{25}$

(b)(ii) [2] $EK = \frac{1}{2}m\langle c^2\rangle = (3/2)kT$

$= 1.5 \times 1.38\times10^{-23} \times 290 = 6.0\times10^{-21}$ J

(b)(iii) [2] For an ideal gas the potential energy is zero, so internal energy = total kinetic energy of all the molecules.

$U = N \times EK = 1.3\times10^{25} \times 6.0\times10^{-21} = 7.8\times10^{4}$ J

(c) [2] A straight line through the origin with positive gradient — internal energy U ∝ T, and at constant pressure V ∝ T, so U is directly proportional to V.

9702/42/M/J/24 Q2 — Ideal gases in thermal equilibrium — temperature & molecule count [10 ma

(a) [1] (If the objects are in thermal contact) there is no net transfer of thermal energy between them.

(b)(i) [3] pV = nRT

$T = (1.20\times10^{5} \times 0.0260) / (0.740 \times 8.31) = 3120 / 6.15 = 507$ K

θ = T − 273 = 507 − 273 = 234 °C (shown)

(b)(ii) [3] The samples are in thermal equilibrium, so the gas in Y has the same temperature as the gas in X: T = 507 K.

pV = NkT

$N = (2.90\times10^{5} \times 0.0430) / (1.38\times10^{-23} \times 507) = 12470 / 6.997\times10^{-21}$

$= 1.78\times10^{24}$

(b)(iii) [3] The mean molecular kinetic energy is the same in both cylinders (equal temperatures), and kinetic energy ∝ mass × mean-square speed, i.e. mean-square speed ∝ 1/m.

Each molecule of gas in X has 4× the mass of a molecule in Y, so $\langle c^2\rangle$ in X is ¼ of $\langle c^2\rangle$ in Y.

Hence the r.m.s. speed in X is half the r.m.s. speed in Y.

9702/41/O/N/24 Q3 — Two ideal gas samples — pressure, mean-square speed & internal energy marks]

(a)(i) [1] The number of particles (atoms or molecules) in one mole of a substance (the number of particles per unit amount of substance).

(a)(ii) [1] $N_a = R/k$

(b)(i) [4] Pressure (pV = NkT): X = NkT/V; Y = 2NkT/2V = NkT/V (the same).

Amount of substance: $X = N/N_a$; $Y = 2N/N_a$.

Mean-square speed ($\frac{1}{2}m\langle c^2\rangle = (3/2)kT$): X = 3kT/m; Y = 3kT/2m.

Internal energy: X = (3/2)NkT; Y = (3/2) × 2N × kT = 3NkT.

(b)(ii) [2] c.r.m.s. = √(3kT/m), so c.r.m.s. ∝ √T: a curve starting at the origin with a positive but continuously decreasing gradient, rising without limit as T increases (it does not return to either axis).

9702/42/O/N/24 Q4 — Kinetic theory — gas pressure & molecular mass from graphs [9 marks]

(a) [3] Any three of: molecules are in (constant) random motion; collisions between molecules are (perfectly) elastic; there are no forces between molecules except during collisions; the volume of the molecules is negligible compared with the volume of the gas; collisions between molecules are instantaneous.

(b) [3] Any three of: molecules collide with the (walls of the) container; the momentum of a molecule changes during a collision with a wall; this change in momentum is caused by a force on the molecule from the wall, so the molecule exerts an equal and opposite force on the wall; many molecules exerting forces across the area of the wall give rise to pressure.

(c) [3] Any three of: both gases are ideal — pV is proportional to T (straight lines through the origin in Fig. 4.2); a molecule of Y has double the mass of a molecule of X (6.6×10^{-27} kg and 3.3×10^{-27}

kg); the sample of Y contains three times the amount of gas / number of molecules as the sample of X (0.81 mol / 4.9×10^{23} against 0.27 mol / 1.6×10^{23}); the mass of gas Y is six times the mass of gas X (3.2×10^{-3} kg against 5.4×10^{-4} kg).

9702/42/F/M/25 Q3 — Thermal equilibrium, specific heat capacity & ideal gas [12 marks]

(a)(i) [2] P and Q are at the same temperature, so there is no net transfer of thermal energy between them.

(a)(ii) [3] Energy is shared between P and Q, so Q = (mcΔT)P + (mcΔT)Q:

$24 \times 10^3 = (0.54 \times 390 \times \Delta T) + (0.37 \times 910 \times \Delta T)$

$24 \times 10^3 = (210.6 + 336.7)\Delta T = 547.3\Delta T$

$\Delta T = 44$ K

(b)(i) [3] Work done = $p\Delta V = (1.6 \times 10^5) \times (0.18 - 0.32)$

$= 1.6 \times 10^5 \times (-0.14) = -2.2 \times 10^4$ J

(The volume decreases, so work is done on the gas.)

(b)(ii) [2] $pV = NkT$, so $N = pV/kT$

$= (1.6 \times 10^5 \times 0.18)/(1.38 \times 10^{-23} \times 273)$

$= 7.6 \times 10^{24}$

(b)(iii) [2] $\frac{1}{2}m\langle c^2\rangle = \frac{1}{2}m(\text{cr.m.s.})^2 = (3/2)kT$, so

cr.m.s. $= \sqrt{(3kT/m)} = \sqrt{[3 \times 1.38 \times 10^{-23} \times (210 + 273)/(4.7 \times 10^{-26})]}$

$= 650$ m s^{-1}

9702/42/M/J/25 Q4 — Ideal gases — kinetic theory & mean kinetic energy [8 marks]

(a)(i) [1] T is the thermodynamic temperature (in kelvin).

(a)(ii) [1] B is the Boltzmann constant (k).

(b)(i) [2] m is the mass of one molecule of the gas, and $\langle c^2\rangle$ is the mean-square speed of the molecules.

(b)(ii) [2] Equating pV from the two equations: $NBT/A = \frac{1}{3}Nm\langle c^2\rangle$. The mean kinetic energy is $Ek = \frac{1}{2}m\langle c^2\rangle$,

$Ek = 3BT/2A$

(c) [2] A line with a positive gradient passing through the origin, becoming a smooth curve with a decreasing positive gradient (since cr.m.s. $\propto \sqrt{T}$).

9702/44/M/J/25 Q3 — Ideal gas — r.m.s. speed, temperature & internal energy [10 marks]

(a) [2] A gas that obeys $pV \propto T$ at all values of p, V and T, where T is the thermodynamic temperature.

(b)(i) [3] $pV = \frac{1}{3}Nm\langle c^2\rangle$ and $Nm/V = \rho$, so $p = \frac{1}{3}\rho\langle c^2\rangle$.

$1.6 \times 10^5 = \frac{1}{3} \times 1.9 \times \langle c^2\rangle$

cr.m.s. $= \sqrt{(3 \times 1.6 \times 10^5/1.9)} = 500$ m s^{-1}

(b)(ii) [2] $\frac{1}{2}m\langle c^2\rangle = (3/2)kT$, so $T = m\langle c^2\rangle/(3k)$

$= 4.7 \times 10^{-26} \times 503^2/(3 \times 1.38 \times 10^{-23})$

$= 290$ K

(c) [3] For an ideal gas the molecular potential energy is zero, so the internal energy is the total kinetic energy: $U = N \times (3/2)kT$.

$U = 6.0 \times 6.02 \times 10^{23} \times (3/2) \times 1.38 \times 10^{-23} \times 290$

$= 21000$ J

9702/41/O/N/25 Q4 — Absolute zero & ideal gas — molecules, moles & r.m.s. speed [10 marks]

(a)(i) [1] temperature = −273.15 °C

(a)(ii) [1] temperature = 0 K

(b)(i) [1] The graph is a straight line through the origin ($pV \propto kT$), so the gas is ideal.

(b)(ii) [2] $pV = NkT$, so $N = pV/kT = 270/(8.0 \times 10^{-21})$

$= 3.4 \times 10^{22}$

(b)(iii) [1] $n = N/N_A = 3.4 \times 10^{22}/(6.02 \times 10^{23})$

$= 0.056$ mol

(c) [4] $\frac{1}{2}m\langle c^2\rangle = (3/2)kT$. When pV = 270 J, $kT = 8.0 \times 10^{-21}$ J:

$\frac{1}{2} \times m \times 1900^2 = 1.5 \times 8.0 \times 10^{-21}$

$m = 6.65 \times 10^{-27}$ kg $= 6.65 \times 10^{-27}/(1.66 \times 10^{-27}) = 4.0$ u

9702/42/O/N/25 Q2 — Newton's law & the kinetic theory of gases [11 marks]

(a) [2] The gravitational force is (directly) proportional to the product of the two masses and inversely proportional to the square of their separation.

(b) [2] Any two of: the molecules are in continuous random motion; the molecules have negligible volume

compared with the volume of the gas; the collisions are (perfectly) elastic; the collisions are instantaneous.

(c)(i) [2] pV = nRT, so p = nRT/V = 0.0160 × 8.31 × 282/(1.87 × 10^{-4})
= 2.01 × 10^{5} Pa

(c)(ii) [2] number of molecules = 0.0160 × 6.02 × 10^{23}.
separation = $\sqrt[3]{(V/N)}$ = $\sqrt[3]{[1.87 \times 10^{-4}/(0.0160 \times 6.02 \times 10^{23})]}$
= 2.7 × 10^{-9} m

(d)(i) [2] F = Gm^2/r^2 = 6.67 × 10^{-11} × $(3.34 \times 10^{-27})^2/(2.7 \times 10^{-9})^2$
= 1.0 × 10^{-46} N

(d)(ii) [1] The gravitational force (≈10^{-46} N) is enormously smaller than the weight of a molecule (≈mg ≈ 10^{-26} N), so the assumption that there are no forces between the molecules is supported.

9702/43/O/N/25 Q4 — Absolute zero & ideal gas — molecules, moles & r.m.s. speed [10 marks]

(a)(i) [1] temperature = −273.15 °C

(a)(ii) [1] temperature = 0 K

(b)(i) [1] The graph is a straight line through the origin (pV ∝ kT), so the gas is ideal.

(b)(ii) [2] pV = NkT, so N = pV/kT = 270/(8.0 × 10^{-21})
= 3.4 × 10^{22}

(b)(iii) [1] n = N/N_A = 3.4 × 10^{22}/(6.02 × 10^{23})
= 0.056 mol

(c) [4] ½m⟨c^2⟩ = (3/2)kT. When pV = 270 J, kT = 8.0 × 10^{-21} J:
½ × m × 1900^2 = 1.5 × 8.0 × 10^{-21}
m = 6.65 × 10^{-27} kg = 6.65 × 10^{-27}/(1.66 × 10^{-27}) = 4.0 u

9702/44/O/N/25 Q2 — Ideal gas — equation of state & r.m.s. speed ratio [7 marks]

(a)(i) [3] p = the pressure of the gas, V = the volume of the gas, N = the number of molecules, k = the Boltzmann constant, T = the thermodynamic temperature.

(a)(ii) [2] pV = NkT = ⅓Nm⟨c^2⟩, so m⟨c^2⟩ = 3kT.
Ek = ½m⟨c^2⟩ = ½ × 3kT = (3/2)kT

(b) [2] At the same temperature ½m⟨c^2⟩ is the same for both gases, so cr.m.s. ∝ 1/√m.
ratio = √(mO/mH) = √(32/2) = 4.0

9702/41/M/J/26 Q2 — Ideal gas — thermometers, r.m.s. speed & temperature [13 marks]

(a) [2] A gas which obeys the equation of state pV = nRT (or pV = NkT) under all conditions, where T is the thermodynamic temperature.

(b)(i) [1] Any one of: it measures absolute (thermodynamic) temperature, unlike a liquid-in-glass thermometer; it has a much greater range of temperature; its measurement does not depend on t properties of a (real) substance.

(b)(ii) [1] Any one of: where the temperature changes rapidly; where the thermometer must have a very sm mass (low heat capacity) so that it does not disturb the surroundings; where a remote reading is required.

(c) [2] □c^2□ = $(357^2 + 424^2 + 509^2 + 286^2 + 435^2)/5$ = 837 327/5 = 167 465
c_0 = √167 465 = 409 m s^{-1}

(d)(i) [2] n = N/N_A = (5.0 × 10^{23})/(6.02 × 10^{23}) = 0.83 mol

(d)(ii) [2] ½mc^2 = (3/2)kT, where c is the r.m.s. speed, so T_0 = mc_0^2/(3k)
= (3.34 × 10^{-27} × 409^2)/(3 × 1.38 × 10^{-23}) = 5.59 × 10^{-22}/4.14 × 10^{-23} = 13.5 K

(d)(iii) [3] A curve starting at the point (T_0, c_0) with a positive gradient at that point. The gradient decreases as T increases, since the r.m.s. speed is proportional to the square root of the thermodynamic temperature. When extrapolated back from T_0, the curve passes through the orig (0, 0).

9702/42/M/J/26 Q4 — Particle in a cubic box — momentum change & pressure derivation [10 ma

(a) [2] Any two of: collisions between particles and with the walls are instantaneous (collision time negligible compared with the time between collisions); collisions are perfectly elastic (no kinetic energy is lost).

(b)(i) [1] The momentum reverses from +mu to −mu:
change in momentum = 2mu

(b)(ii) [1] The particle travels a distance 2L (there and back) between successive collisions with the same

face:

time = 2L/u

(b)(iii) [1] F = Δp/t = 2mu/(2L/u)

= mu²/L

(b)(iv) [2] By Newton's third law the force exerted on the face by the particle is equal (and opposite) to the force exerted on the particle by the face. Pressure = force/area:

p = F/A = (mu²/L)/L² = mu²/L³

(c) [3] Three modifications: the volume of the cube L³ is the volume V of the gas, so pV = mu²; particles move in three (all) dimensions, so only one-third of the mean-square speed acts along any one axis — u² becomes ⅓⟨c²⟩; with N particles the pressure is N times that due to a single particle:

pV = ⅓Nm⟨c²⟩

9702/43/M/J/26 Q2 — Ideal gas — r.m.s. speed, amount of substance & temperature [13 marks]

(a) [2] A gas that obeys the relationship pV ∝ T (i.e. pV = nRT or pV = NkT) under all conditions, where T is the thermodynamic (Kelvin) temperature.

(b)(i) [1] It measures the absolute (thermodynamic) temperature directly — its scale does not depend on th properties of any particular substance, and it can measure a much greater range of temperatures than a liquid-in-glass thermometer.

(b)(ii) [1] Any one of: measuring rapidly changing temperatures; measuring the temperature of a very smal mass (amount) of substance; taking a remote reading — the thermocouple has a very small therm capacity and responds quickly.

(c) [2] □c²□ = (357² + 424² + 509² + 286² + 435²)/5 = 837327/5 = 167465

c0 = √167465 = 409 m s⁻¹ (to 3 s.f.)

(d)(i) [2] n = N/NA = 5.0×10²³ / 6.02×10²³ = 0.83 mol

(d)(ii) [2] ½m□c²□ = (3/2)kT, so T0 = m□c²□/(3k)

T0 = (3.34×10⁻²⁷ × 409²)/(3 × 1.38×10⁻²³) = 5.59×10⁻²²/4.14×10⁻²³

T0 = 13.5 K

(d)(iii) [3] A curve starting at the point (T0, c0) with a positive gradient at that point.

The gradient is positive but decreasing as T increases (the line becomes almost horizontal, but never of negative gradient).

When extrapolated back from T = T0 towards T = 0, the line passes through the origin (0, 0) — consistent with c ∝ √T.

16 Thermodynamics

9702/42/F/M/24 Q2 — Ideal gas expansion — work done, temperature & specific heat capacity [1 marks]

(a) [2] Internal energy is the sum of the randomly distributed kinetic energies and the potential energies of the molecules. For an ideal gas there are no intermolecular forces, so there is no potential energy — the internal energy is the sum of the kinetic energies of the molecules only.

(b)(i) [2] $W = p\Delta V = 1.01\times10^{5} \times 5.20\times10^{-5} = 5.25$ J

(b)(ii) [2] At constant pressure V/T = constant. Final volume = $1.24\times10^{-4} + 5.20\times10^{-5} = 1.76\times10^{-4}$ m³

$T = 293 \times (1.76\times10^{-4}/1.24\times10^{-4}) = 416$ K

(b)(iii) [2] $c = E/(m\Delta T) = 960/(0.016\times(420-293)) = 470$ J kg^{-1} K^{-1} (the mark scheme also accepts 490 J kg^{-1} K^{-1} using the unrounded final temperature of 416 K)

(c) [3] With the piston fixed there is no change in volume, so $W = 0$. The same temperature rise gives the same ΔU (the internal energy of an ideal gas depends only on temperature). From the first law $\Delta U = q + W$, less energy by heating q is needed than in (b) (where the gas also did work), so $c = q/(m\Delta T)$ is less than in (b)(iii).

9702/42/M/J/24 Q3 — Internal energy — heating a gas and stretching a wire [8 marks]

(a) [2] The internal energy of a system is the sum of the kinetic energies and potential energies of its molecules — the total energy of the random motion of the particles of the system.

(b)(i) [3] Heated at constant volume: the molecular separation does not change, so the molecular potential energy is unchanged.

The temperature increases, so the molecular kinetic energy increases.

Kinetic energy increases and potential energy is unchanged, so the internal energy increases.

(b)(ii) [3] Stretched within its elastic limit at constant temperature: the temperature is constant, so the molecular kinetic energy is unchanged.

The molecules are pulled further apart, so the molecular potential energy increases.

Potential energy increases and kinetic energy is unchanged, so the internal energy increases.

9702/41/M/J/25 Q4 — Internal energy & the first law of thermodynamics [8 marks]

(a)(i) [2] The sum of the (randomly distributed) kinetic and potential energies of the particles of the system — the total energy of the random motion of its particles.

(a)(ii) [2] For an ideal gas there are no intermolecular forces, so the potential energy is zero and the internal energy equals the total kinetic energy of the molecules. The average kinetic energy of the molecules is proportional to the (thermodynamic) temperature, so the internal energy is directly proportional to the thermodynamic temperature.

(b) [4] Compression (T → 3T): internal energy rises threefold, so $\Delta U = +2U$; work done on the gas = +W. From the first law $\Delta U = q + W$, so thermal energy supplied $q = 2U - W$.

Cooling at constant volume (3T → 2T): work done = 0; the internal energy falls from 3U to 2U, so $\Delta U = -U$; hence thermal energy supplied $q = \Delta U = -U$.

9702/43/M/J/25 Q4 — Internal energy & the first law of thermodynamics [8 marks]

(a)(i) [2] The sum of the (randomly distributed) kinetic and potential energies of the particles of the system — the total energy of the random motion of its particles.

(a)(ii) [2] For an ideal gas there are no intermolecular forces, so the potential energy is zero and the internal energy equals the total kinetic energy of the molecules. The average kinetic energy of the molecules is proportional to the (thermodynamic) temperature, so the internal energy is directly proportional to the thermodynamic temperature.

(b) [4] Compression (T → 3T): internal energy rises threefold, so $\Delta U = +2U$; work done on the gas = +W. From the first law $\Delta U = q + W$, so thermal energy supplied $q = 2U - W$.

Cooling at constant volume (3T → 2T): work done = 0; the internal energy falls from 3U to 2U, so $\Delta U = -U$; hence thermal energy supplied $q = \Delta U = -U$.

9702/41/O/N/25 Q2 — The first law of thermodynamics — pumping, boiling & vaporisation [8 ma

(a) [2] The internal energy can be changed by (1) work done on or by the system, and (2) thermal energy supplied to or removed from the system.

(b)(i) [3] Pumping quickly gives no time for thermal energy to be transferred to or from the gas, while

work is done on the gas to compress it (decrease its volume). From the first law $\Delta U = q + W$ (with $q = 0$), the internal energy increases, so the temperature rises.

(b)(ii) [3] During vaporisation the molecular separation increases, so the heating increases the potential energy of the molecules while their kinetic energy is unchanged. Since temperature depends on the kinetic energy, the temperature stays at 100 °C.

9702/42/O/N/25 Q4 — The first law — an ideal gas thermodynamic cycle [10 marks]

(a) [2] The increase in internal energy = work done on the system + thermal energy transferred to the system by heating ($\Delta U = q + W$).

(b)(i) [1] $U = (3/2)pV$

(b)(ii) [2] $pV = NkT$ (k = the Boltzmann constant), so $U = (3/2)NkT$

(c)(i) [1] $W = +8XY$ (compression — work done on the gas).

(c)(ii) [1] $W = -20XY$ (expansion — work done by the gas).

(d) [3] The work done during BC and DA is zero (constant volume). Over the complete cycle the change in internal energy is zero (the gas returns to A). From the first law, the net thermal energy supplied $Q = -$(net work done on the gas) $= 20XY - 8XY$

$= +12XY$

9702/43/O/N/25 Q2 — The first law of thermodynamics — pumping, boiling & vaporisation [8 ma

(a) [2] The internal energy can be changed by (1) work done on or by the system, and (2) thermal energy supplied to or removed from the system.

(b)(i) [3] Pumping quickly gives no time for thermal energy to be transferred to or from the gas, while work is done on the gas to compress it (decrease its volume). From the first law $\Delta U = q + W$ (with $q = 0$), the internal energy increases, so the temperature rises.

(b)(ii) [3] During vaporisation the molecular separation increases, so the heating increases the potential energy of the molecules while their kinetic energy is unchanged. Since temperature depends on the kinetic energy, the temperature stays at 100 °C.

9702/44/O/N/25 Q3 — Internal energy of an ideal gas — changes X and Y [9 marks]

(a) [2] For an ideal gas the internal energy is the total kinetic energy of the random motion of the molecules, because the potential energy is zero (there are no intermolecular forces).

(b)(i) [2] $U = (3/2)NkT = (3/2)pV = (3/2) \times 2.0 \times 10^5 \times 0.016$

$= 4800$ J

(b)(ii) [1] At constant volume $p \propto T$, so doubling the pressure doubles the temperature: $T = 400 \times 2 = 800$

(b)(iii) [1] At constant temperature pV is constant, so halving the pressure doubles the volume: $V = 0.016 \times 2 = 0.032$ m³

(b)(iv) [3] Change X: a straight vertical line from A to B at $(0.016, 4.0 \times 10^5)$, labelled X (constant volume).

Change Y: a curve with a continuously decreasing negative gradient from B to C at $(0.032, 2.0 \times 10^5)$, labelled Y (isothermal).

9702/41/M/J/26 Q3 — Internal energy of an ideal gas — first law & p–V cycle [10 marks]

(a) [2] The internal energy of a system is the sum of the potential energy and the kinetic energy of its particles — the total energy of the random motion of the particles.

(b)(i) [2] $pV = NkT \rightarrow V = NkT/p$

$= (8.7 \times 10^{23} \times 1.38 \times 10^{-23} \times 300)/(2.0 \times 10^5) = 3602/2.0 \times 10^5 = 0.018$ m³

(b)(ii) [3] For an ideal gas there are no intermolecular forces, so the potential energy of the molecules is zero — the internal energy is the total kinetic energy of the molecules.

$U = (3/2)NkT = 1.5 \times 8.7 \times 10^{23} \times 1.38 \times 10^{-23} \times 300 = 5400$ J

(c) [3] Any three of:

AB: the internal energy does not change, so the work done on the gas equals the thermal energy transferred from it.

BC: the volume is constant, so no work is done on or by the gas; the internal energy is halved (decreases by 2.7 kJ), and this decrease in internal energy equals the thermal energy removed.

CA: the internal energy is doubled (from 2.7 kJ to 5.4 kJ); the work done by the gas is 1.8 kJ and the thermal energy transferred to the gas is 4.5 kJ.

Whole cycle: there is no overall change in internal energy; overall, work is done on the gas and an equal amount of thermal energy is removed from it (net work done = net thermal energy

transferred).

9702/42/M/J/26 Q3 — Cooling gas at constant pressure — internal energy & specific heat capacit [14 marks]

(a) [2] The specific heat capacity of a substance is the thermal energy per unit mass per unit change in temperature.

(b)(i) [2] $W = p\Delta V = 1.01\times10^5 \times (1.00 - 0.80)\times0.0243$
$= 491$ J

(b)(ii) [1] The volume of the gas decreases (the gas is compressed), so the work done on the gas is positive.

(b)(iii) [2] For an ideal gas the internal energy is proportional to the thermodynamic temperature ($U = \frac{3}{2}NkT$). At constant pressure the temperature is proportional to the volume, so the internal energy falls to 80% of its initial value:
$\Delta U = -0.20\times3676 = -735$ J

(b)(iv) [2] First law: $\Delta U = Q + W$ (W is the work done on the gas)
$-735 = Q + 491$
$Q = -1226$ J (thermal energy is removed from the gas)

(b)(v) [2] $\Delta T = 0.20\times296 = 59$ K
$c = Q/(m\Delta T) = 1226/(6.00\times10^{23}\times6.64\times10^{-27}\times0.20\times296)$
$= 5200$ J kg^{-1} K^{-1}

(c) [3] The same temperature change gives the same change in internal energy as in (b). At constant volume no work is done on or by the gas ($W = 0$), so from $\Delta U = Q + W$ only 735 J of thermal energy needs to be supplied — less than the 1226 J removed in (b). Hence $c = Q/(m\Delta T)$ is less than the value in (b)(v).

9702/43/M/J/26 Q3 — Internal energy — p–V cycle of an ideal gas [10 marks]

(a) [2] The internal energy of a system is the sum of the kinetic energies and potential energies of its molecules — the total energy of the random motion of its particles.

(b)(i) [2] $pV = NkT$
$V = (8.7\times10^{23} \times 1.38\times10^{-23} \times 300)/(2.0\times10^5) = 3602/2.0\times10^5 = 0.018$ m^3 (shown)

(b)(ii) [3] For an ideal gas there are no intermolecular forces, so the potential energy of the molecules is zero — the internal energy is entirely the total kinetic energy of the molecules.
$U = (3/2)NkT = 1.5 \times 8.7\times10^{23} \times 1.38\times10^{-23} \times 300$
$U = 5.4\times10^3$ J

(c) [3] Any three conclusions (each must be related to a specific change or to the whole cycle):
AB (isothermal): no change in internal energy (pV is constant); work done on the gas equals the thermal energy transferred from it.
BC (constant volume): no work is done; the internal energy halves to 2.7 kJ; the decrease in internal energy equals the thermal energy removed (2.7 kJ).
CA (constant pressure): the internal energy doubles (increases by 2.7 kJ to 5.4 kJ); work done by the gas = $p\Delta V = 2.0\times10^5 \times 0.009 = 1.8$ kJ; thermal energy transferred to the gas = 4.5 kJ.
Complete cycle: no overall change in internal energy; net work is done on the gas and thermal energy is removed from it (net work done = net thermal energy transferred).

9702/44/M/J/26 Q4 — Ideal gas internal energy — molecular kinetic energy & first law [13 marks

(a) [2] The increase in internal energy of a system is equal to the work done on the system plus the energy transferred to the system by heating: $\Delta U = q + W$.

(b)(i) [2] $pV = NkT \rightarrow V = NkT/p$
$V = 3.58\times10^{24}\times1.38\times10^{-23}\times298/(1.84\times10^5) = 1.47\times10^4/1.84\times10^5$
$= 0.0800$ m^3

(b)(ii) [1] $n = N/N_a = 3.58\times10^{24}/6.02\times10^{23} = 5.95$ mol

(b)(iii) [2] $E_K = (3/2)kT = 1.5\times1.38\times10^{-23}\times298 = 6.17\times10^{-21}$ J

(b)(iv) [1] An ideal gas has no intermolecular forces, so the molecules have no potential energy — only kinetic energy.

(b)(v) [1] $U = NE_K = 3.58\times10^{24}\times6.17\times10^{-21} = 2.21\times10^4$ J

(c)(i) [2] During AB the volume is constant, so no work is done ($W = 0$); by the first law $Q = \Delta U$. For an ideal gas $U \propto pV$ and V is constant, so $\Delta U/U_a = \Delta p/p$:

$Q = (5.84 - 1.84)/1.84 \times U_a = 2.2U_a$ $(50U_a/23)$

(c)(ii) [2] During BC there is no transfer of thermal energy (adiabatic), so $W = \Delta U$. At C, $U_C \propto pV = 1.84 \times 10^5 \times 0.160$, which is twice pV at A, so $U_C = 2U_a$:

$W = U_C - U_B = 2U_a - (U_a + 2.2U_a) = -1.2U_a$ $(-27U_a/23)$ — $1.2U_a$ of work is done by the gas.

17 Oscillations

9702/42/F/M/24 Q3 — Object on an oscillating platform — angular frequency & amplitude [7 mar

(a) [3] Distance travelled per oscillation = $4x_0 \rightarrow x_0 = 14\times10^{-3}/4 = 3.5\times10^{-3}$ m

$E = \frac{1}{2}m\omega^2x_0^2 \rightarrow \omega^2 = 2E/(mx_0^2) = 2\times2.2\times10^{-4}/(0.024\times(3.5\times10^{-3})^2) = 1.50\times10^3$

$\omega = 39$ rad s^{-1}

(b)(i) [2] The object loses contact when the maximum downward acceleration of the platform, ω^2x_0, exceed g:

$x_0 = g/\omega^2 = 9.81/39^2 = 6.4\times10^{-3}$ m

(b)(ii) [2] At the top of the oscillation — there the acceleration of the platform is a maximum downwards (a = $-\omega^2x$, magnitude ω^2x_0). The object's maximum possible downward acceleration is g = 9.81 m s so it loses contact when ω^2x_0 exceeds 9.81 m s^{-2}, i.e. first at the top of the oscillation.

9702/41/M/J/24 Q4 — Resonance — damping & forced oscillations of a ball [8 marks]

(a) [2] Resonance is the phenomenon in which an oscillating system has its amplitude a maximum, and this occurs when the driving frequency equals the natural frequency of the system.

(b)(i) [1] Light damping.

(b)(ii) [2] The oscillations of the ball lose energy because resistive forces (air resistance / friction in the string) act on the ball and do work against its motion, so the amplitude decreases gradually with time.

(b)(iii) [1] Period T = 0.25 s (time between successive maxima of Fig. 4.2).

$f = 1/T = 1/0.25 = 4.0$ Hz

(c) [2] A resonance curve: the amplitude rises from zero at f = 0, reaches a single maximum at f = 4.0 Hz (the natural frequency of the ball), and then falls again as f increases towards 10 Hz.

9702/42/M/J/24 Q4 — Block oscillating on a spring — simple harmonic motion & energy [9 marks

(a) [2] Fig. 4.2 is a straight line through the origin, showing that a is proportional to x; its negative gradient shows that a and x are always in opposite directions, i.e. $a = -\omega^2x$.

(b)(i) [1] $a = -\omega^2x$ with x0 = 3Y and a_max = 2A:

$2A = \omega^2(3Y)$

$\omega = \sqrt{(2A/3Y)}$

(b)(ii) [2] $v0 = \omega x0 = \sqrt{(2A/3Y)} \times 3Y$

$= \sqrt{(6AY)}$

(b)(iii) [2] $E = \frac{1}{2}m\omega^2x0^2 = \frac{1}{2}m \times (2A/3Y) \times (3Y)^2$

$= \frac{1}{2}m \times 6AY = 3mAY$

(c) [2] $\omega = 2\pi/T = 2\pi/0.75 = 8.4$ rad s^{-1}; amplitude x0 = 3Y = 1.8 cm

$x = x0 \sin \omega t$

$x = 1.8 \sin (8.4t)$ (x in cm, t in s)

9702/41/O/N/24 Q4 — Block oscillating on a spring — amplitude, angular frequency & sketch [11 marks]

(a) [2] Motion in which the acceleration is directly proportional to the displacement from a fixed point (the equilibrium position) and is always directed towards that fixed point (opposite in direction to the displacement).

(b)(i) [1] From Fig. 4.2, the turning points (v = 0) are at h = 3.5 cm and h = 9.5 cm.

amplitude = (9.5 − 3.5)/2 = 3.0 cm

(b)(ii) [2] $v_{max} = v_0 = \omega x_0 \rightarrow \omega = v_0/x_0$

$\omega = 9.5/3.0 = 3.2$ rad s^{-1}

(b)(iii) [2] $T = 2\pi/\omega = 2\pi/3.2 = 2.0$ s

(b)(iv) [4] A sinusoidal curve starting at a minimum (h = 3.5 cm) at t = 0, since the block is released from rest at its lowest point. Three complete cycles from t = 0 to t = 6.0 s (period 2.0 s), with all peaks at h = 9.5 cm and all troughs at h = 3.5 cm.

9702/42/O/N/24 Q5 — Pendulum in simple harmonic motion — energy & EK sketch [9 marks]

(a) [1] An arrow drawn from the centre of the sphere, perpendicular to the string, pointing left and down — the resultant (restoring) force is directed towards the equilibrium position.

(b)(i) [1] amplitude = 0.016 m (maximum displacement, read from Fig. 5.2)

(b)(ii) [2] T = 0.40 s (from Fig. 5.2)
$\omega = 2\pi/T = 2\pi/0.40$
$= 16$ rad s^{-1}
(b)(iii) [2] $E = \frac{1}{2}m\omega^2x_0^2 = \frac{1}{2} \times 0.15 \times 15.7^2 \times 0.016^2$
$= 4.7\times10^{-3}$ J
(c) [3] A dome-shaped curve starting and ending on the x-axis at x = ±0.016 m, rising to a maximum of 4.7×10^{-3} J at x = 0 — kinetic energy is zero at the extremes of the oscillation and maximum at the equilibrium position.

9702/42/F/M/25 Q4 — Piezoelectric crystal — simple harmonic motion & ultrasound [12 marks]

(a) [2] $\omega = 2\pi/T = 2\pi/(0.15 \times 10^{-6})$
$= 4.2 \times 10^7$ rad s^{-1}
(b) [2] $a_0 = \omega^2x_0 = (4.2 \times 10^7)^2 \times 40 \times 10^{-6}$
$= 7.1 \times 10^{10}$ m s^{-2}
(c) [3] $E = \frac{1}{2}m\omega^2x_0^2 = \frac{1}{2} \times 2.4 \times 10^{-4} \times (4.2 \times 10^7)^2 \times (40 \times 10^{-6})^2$
$= 340$ J
(d)(i) [2] Apply an alternating p.d. across the crystal. The p.d. causes the crystal to distort (change shape) at the frequency of the alternating p.d., so the surface vibrates — and conversely an oscillating crystal generates an alternating e.m.f., so it can be used to generate ultrasound.
(d)(ii) [3] $Z = \rho c$, so $Z_{muscle} = 1100 \times 1600 = 1.76 \times 10^6$ and $Z_{bone} = 1900 \times 4100 = 7.79 \times 10^6$ kg m^{-2}
Intensity reflection coefficient $= (Z_1 - Z_2)^2/(Z_1 + Z_2)^2$
$= (7.79 - 1.76)^2/(7.79 + 1.76)^2 = 0.40$ (40%)
so percentage transmitted = 60%

9702/41/M/J/25 Q5 — Floating block — simple harmonic motion & energy [8 marks]

(a)(i) [1] The oscillation extends from h = 0.8 m to h = 2.0 m about the equilibrium at h = 1.4 m, so the amplitude = 0.60 m.
(a)(ii) [1] The line is a straight line through the origin, showing $a \propto -$(displacement), so the oscillations are simple harmonic.
(b) [3] Any three of: the mean/equilibrium position is at h = 1.4 m; the total energy of the oscillations is 9.0 J; the angular frequency is 1.2 rad s^{-1} (period 5.1 s, or frequency 0.19 Hz); the maximum speed is 0.73 m s^{-1}; the mass of the block is 33 kg.
(c) [3] A U-shaped curve resting on the h-axis (minimum at EP = 0), drawn between h = 0.8 m and h = 2.0 m with the minimum at h = 1.4 m and both end-points at EP = 9.0 J (since EP = total energy − EK).

9702/42/M/J/25 Q5 — Simple harmonic motion — displacement, velocity & phase [9 marks]

(a) [2] Motion in which the acceleration is (directly) proportional to the displacement and is always directed towards a fixed point (in the opposite direction to the displacement).
(b)(i) [1] period $= 2\pi/\omega = 2\pi/16$
$= 0.39$ s
(b)(ii) [2] $v_0 = \omega x_0$, so $x_0 = v_0/\omega = 0.56/16$
$= 0.035$ m
(b)(iii) [1] $v = \pm\omega\sqrt{(x_0^2 - x^2)} = \pm16\sqrt{(0.035^2 - x^2)}$
(b)(iv) [3] A closed loop (ellipse) surrounding the origin, crossing the x-axis (v = 0) at x = ±3.5 cm and crossing the v-axis (x = 0) at v = ±0.56 m s^{-1}.

9702/43/M/J/25 Q5 — Floating block — simple harmonic motion & energy [8 marks]

(a)(i) [1] The oscillation extends from h = 0.8 m to h = 2.0 m about the equilibrium at h = 1.4 m, so the amplitude = 0.60 m.
(a)(ii) [1] The line is a straight line through the origin, showing $a \propto -$(displacement), so the oscillations are simple harmonic.
(b) [3] Any three of: the mean/equilibrium position is at h = 1.4 m; the total energy of the oscillations is 9.0 J; the angular frequency is 1.2 rad s^{-1} (period 5.1 s, or frequency 0.19 Hz); the maximum speed is 0.73 m s^{-1}; the mass of the block is 33 kg.
(c) [3] A U-shaped curve resting on the h-axis (minimum at EP = 0), drawn between h = 0.8 m and h = 2.0 m with the minimum at h = 1.4 m and both end-points at EP = 9.0 J (since EP = total energy − EK).

9702/44/M/J/25 Q4 — Simple pendulum — frequency & maximum displacement [8 marks]

(a) [2] Motion in which the acceleration is (directly) proportional to the displacement and is always directed towards a fixed point (in the opposite direction to the displacement).

(b)(i) [1] The sphere passes through the equilibrium position in the same direction at t_1 and t_5 (or t_3 and t_7).

(b)(ii) [2] The interval $t_1 \rightarrow t_6$ is 1.25 periods, so T = 2.2/1.25 = 1.76 s.

f = 1/T = 1.25/2.2 = 0.57 Hz

(c) [3] $v_0 = \omega x_0$ with $\omega = 2\pi f$, so $x_0 = v_0/(2\pi f) = 0.080/(2\pi \times 0.57)$

= 0.022 m

9702/44/M/J/25 Q7 — Damped oscillations — magnet in a coil & electromagnetic induction [8 ma

(a)(i) [2] Damping is the loss of energy from the oscillations due to resistive forces.

(a)(ii) [1] The amplitude of the oscillations decreases gradually (the oscillations continue for several periods).

(a)(iii) [3] The oscillating magnet cuts magnetic flux, inducing an e.m.f. in the coil. This drives a current through the resistor, which dissipates thermal energy. By conservation of energy this thermal energy comes from the energy of the oscillations, so the oscillations are damped.

(b) [2] For a given induced e.m.f. the current is smaller, so thermal energy is dissipated at a lower rate (the resistive force is smaller). The oscillations are therefore less damped — the amplitude decreases more slowly and they continue for longer.

9702/42/O/N/25 Q5 — Simple harmonic motion & heavy damping — steel ball in oil [10 marks]

(a)(i) [2] The graph is a straight line through the origin, showing a is proportional to x, and its negative gradient shows a is always in the opposite direction to x — so the oscillations are simple harmonic.

(a)(ii) [3] $a_0 = \omega^2 x_0$ and $\omega = 2\pi/T$, so $T = 2\pi\sqrt{(x_0/a_0)} = 2\pi\sqrt{(1.2/13)}$

= 1.9 s

(b)(i) [2] Damping is the loss of energy from the oscillations due to resistive forces.

(b)(ii) [3] A line starting at x = ±1.2 cm at t = 0 (entirely above or below the t-axis), with both the magnitude of x and the magnitude of the gradient continuously decreasing — heavy damping, so ball returns to equilibrium without oscillating.

9702/44/O/N/25 Q4 — Simple harmonic motion — frequency, energy & interchanges [10 marks]

(a) [1] The number of oscillations per unit time.

(b)(i) [1] The kinetic energy reaches its maximum/minimum/zero twice in a cycle, so one complete cycle takes 0.80 s.

(b)(ii) [2] $\omega = 2\pi/T = 2\pi/0.80$

= 7.9 rad s^{-1} (or $\omega = \sqrt{(a_0/x_0)} = \sqrt{(1.0/0.016)}$)

(b)(iii) [3] Any three of: the oscillations are simple harmonic; the amplitude = 0.016 m; the maximum speed 0.13 m s^{-1}; the total energy = 7.0×10^{-4} J; the mass of the object = 0.087 kg; the maximum momentum = 0.011 kg m s^{-1}.

(b)(iv) [3] The kinetic energy is a maximum at zero displacement and zero at maximum displacement, while potential energy is zero at zero displacement and a maximum at maximum displacement. The su the kinetic and potential energies is constant (total energy).

9702/42/M/J/26 Q5 — Block oscillating on a spring — amplitude & kinetic energy [8 marks]

(a)(i) [1] $\omega = 2\pi/T = 2\pi/1.7$

= 3.7 rad s^{-1}

(a)(ii) [1] The amplitude is the magnitude of the maximum displacement of the block from its equilibrium (mean) position.

(a)(iii) [3] $E = \frac{1}{2}m\omega^2 x_0^2$

$13\times10^{-6} = \frac{1}{2}\times27\times10^{-3}\times3.7^2\times x_0^2$

$x_0^2 = 2\times13\times10^{-6}/(0.027\times3.7^2) = 7.03\times10^{-5}$

$x_0 = 8.4\times10^{-3}$ m

(b) [3] Sketch a dome-shaped curve, symmetrical about the EK-axis, with a single peak of 13 μJ at x = 0, falling to EK = 0 at x = ±8.4×10^{-3} m (the amplitude). Add a consistent x-axis scale with at least one marking in each half (e.g. ±8 and ±4, in units of 10^{-3} m), corresponding to the end

points.

9702/44/M/J/26 Q5 — Ball on a spring — velocity–displacement graph & resonance [9 marks]

(a)(i) [3] Draw a smooth closed loop (ellipse) enclosing the origin, cutting the x-axis at x = ±2.5 cm (and nowhere further out) and the v-axis at v = ±4.4 cm s^{-1} (and nowhere further out). The v–x graph for s.h.m. is an ellipse, $v = \pm\omega\sqrt{(x_0^2 - x^2)}$, not straight lines between the intercepts.

(a)(ii) [2] $v_0 = \omega x_0$ and $\omega = 2\pi f$:

$f = v_0/(2\pi x_0) = 4.4/(2\pi \times 2.5) = 4.4/15.7$

$= 0.28$ Hz

(b)(i) [2] Resonance is the oscillation of an object at maximum amplitude, which occurs when the driving frequency equals the natural frequency of the oscillating system.

(b)(ii) [2] Draw a curve rising continuously from f = 0 (amplitude never zero when f > 0) to a single peak at f = 0.28 Hz, then falling continuously to f = 0.40 Hz.

18 Electric fields

9702/41/M/J/24 Q5 — Parallel plates — electric field & undeflected electron in crossed fields [13 marks]

(a) [2] The electric field strength at a point is the force per unit positive charge exerted on a small test charge placed at that point.

(b)(i) [2] Four straight, parallel lines between the plates, approximately evenly spaced, running from the +430 V plate towards the 0 V plate, with arrows pointing downwards.

(b)(ii) [2] $E = V/d = 430 / 0.067 = 6.4\times10^{3} \rightarrow 6.4\times10^{3}$ N C^{-1}

(b)(iii) [2] A smooth curve within the plates, bending upwards towards the +430 V plate (the electron is deflected towards the positive plate), with straight lines outside the plates continuing along the tangent to the curve at each edge.

(c)(i) [1] Into the page (perpendicular to both the electron velocity and the electric field), so that the magnetic force on the electron is opposite to the electric force.

(c)(ii) [2] The electric force on the electron is upwards and the magnetic force is downwards. When their magnitudes are equal the forces are equal and opposite, so the resultant force on the electron is zero and its path is undeviated.

(c)(iii) [2] $Eq = Bqv \rightarrow B = E/v = (6.4\times10^{3}) / (2.6\times10^{7}) = 2.5\times10^{-4}$ T

9702/42/M/J/24 Q5 — Electric potential of two charged spheres — proton at rest [7 marks]

(a) [2] The work done per unit positive charge
in bringing a small test charge from infinity to the point.

(b) [3] Any three of:
the radius of sphere X is 0.30 m ($V \rightarrow \infty$ as $x \rightarrow 0.30$ m)
the radius of sphere Y is 0.10 m ($V \rightarrow \infty$ as $x \rightarrow 1.10$ m)
both spheres are positively charged (V is positive throughout)
the charges on the spheres have the same magnitude (V is a minimum at the midpoint x = 0.60 m)

(c) [2] The proton remains at rest.
At x = 0.60 m the electric forces from the two spheres are equal and opposite (equal charges at equal distances), so the resultant force on the proton is zero — equivalently, its electric potential energy is already at its minimum there.

9702/41/O/N/24 Q5 — Two charged spheres — electric potential & resultant field strength [10 m

(a) [2] The electric field strength at a point is equal to the (negative) potential gradient at that point: $E = -\Delta V/\Delta d$, the minus sign showing the field is directed from higher to lower potential.

(b) [3] For the total potential at P to be zero, the two potentials must be of opposite sign, so the charges must have opposite signs. For the resultant field at P to be zero, the two fields must be equal in magnitude and opposite in direction, which (with P between the spheres) requires the charges to have the same sign. The charges cannot be both the same and opposite in sign at the same time, so the potential and the field cannot both be zero at P.

(c)(i) [2] $V_X = Q/4\pi\varepsilon_0 x$ and $V_Y = 2Q/4\pi\varepsilon_0 y$ (magnitudes).
The total potential is zero, so $V_X = V_Y$:
$Q/4\pi\varepsilon_0 x = 2Q/4\pi\varepsilon_0 y$, giving $y = 2x$

(c)(ii) [1] $E_X = Q/4\pi\varepsilon_0 x^2$

(c)(iii) [2] $E_Y = 2Q/4\pi\varepsilon_0 (2x)^2 = Q/8\pi\varepsilon_0 x^2$
The charges have opposite signs, so at P (between the spheres) the two fields act in the same direction and their magnitudes add:
$E = Q/4\pi\varepsilon_0 x^2 + Q/8\pi\varepsilon_0 x^2 = 3Q/8\pi\varepsilon_0 x^2$

9702/42/O/N/24 Q6 — Charged hollow sphere — Coulomb's law & field strength [10 marks]

(a) [2] The (electric) force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of their separation.

(b) [3] At least four straight, radial lines from the surface of the sphere, approximately equally spaced, with arrows pointing away from the sphere (field lines leave the positive charge and extend to infinity).

(c)(i) [1] The field is zero inside the sphere, so the plotted curve begins at the surface: radius = 3.2 cm.

(c)(ii) [3] $E = Q/(4\pi\varepsilon_0 x^2) \rightarrow Q = 4\pi\varepsilon_0 x^2 E$
$= 2.2\times10^5 \times 4\pi \times 8.85\times10^{-12} \times 0.032^2$
$= 2.5\times10^{-8}$ C (with $E = 2.2\times10^5$ N C^{-1} read from Fig. 6.2 at the surface)

(c)(iii) [1] Any one of: the (positive) charge is spread all the way around the surface; a charge placed inside the sphere is pulled equally in all directions, so the fields cancel; if the field were not zero, the charges would move until it was; field lines go from positive to negative charge and there are no negative charges inside the sphere.

9702/41/M/J/25 Q2 — Helium atom model — Coulomb's law, electron orbits & field strength [11 r

(a) [2] The (electric) force between two point charges is (directly) proportional to the product of their charges and inversely proportional to the square of their separation.

(b)(i) [1] The helium nucleus contains two protons, so its charge is +2e.

(b)(ii) [1] $F = Qq/(4\pi\varepsilon_0 r^2) = 2 \times (1.60 \times 10^{-19})^2/[4\pi \times 8.85 \times 10^{-12} \times (170 \times 10^{-12})^2]$
$= 1.6 \times 10^{-8}$ N

(c)(i) [2] The electric force provides the centripetal force: $F = mv^2/r$, so
$v = \sqrt{(Fr/m)} = \sqrt{(1.6 \times 10^{-8} \times 170 \times 10^{-12} / 9.11 \times 10^{-31})}$
$= 1.7 \times 10^6$ m s^{-1}

(c)(ii) [2] $v = 2\pi r/T$, so $T = 2\pi r/v = 2\pi \times 170 \times 10^{-12} / (1.7 \times 10^6)$
$= 6.2 \times 10^{-16}$ s
(Equivalently $F = 4\pi^2 mr/T^2$ gives the same period.)

(d)(i) [2] $E \propto Q/r^2$. The other electron has charge e at a distance of 340 pm, and the nucleus has charge 2e at 170 pm, so
ratio $= [e/(340 \times 10^{-12})^2]/[2e/(170 \times 10^{-12})^2] = (170)^2/(2 \times 340)^2 = 0.13$

(d)(ii) [1] The other electron exerts a small repulsive force, so the resultant inward force is slightly less than 1.6×10^{-8} N — hence the speed of the electron is slightly lower (or its period slightly greater).

9702/42/M/J/25 Q6 — Electric fields & X-rays — an accelerated electron [11 marks]

(a) [1] Plate X is negative (−) and plate Y is positive (+) (the electron accelerates towards the positive plate).

(b)(i) [2] $E = V/x = 58 \times 10^3/0.041$
$= 1.4 \times 10^6$ N C^{-1} (equivalent to V m^{-1})

(b)(ii) [2] $ma = eE$, so $a = eE/m = (1.60 \times 10^{-19} \times 1.41 \times 10^6)/9.11 \times 10^{-31}$
$= 2.5 \times 10^{17}$ m s^{-2}

(c)(i) [3] The electron gains kinetic energy eV, which is converted to a photon at the minimum wavelength: $eV = hc/\lambda$.
$\lambda = hc/eV = (6.63 \times 10^{-34} \times 3.00 \times 10^8)/(1.60 \times 10^{-19} \times 58 \times 10^3)$
$= 2.1 \times 10^{-11}$ m = 21 pm

(c)(ii) [1] X-rays.

(c)(iii) [2] The X-rays are passed into the structure and the transmitted waves are detected. Different parts of the structure absorb different fractions of the energy, and the differences in the detected/transmitted intensities are used to form the image.

9702/43/M/J/25 Q2 — Helium atom model — Coulomb's law, electron orbits & field strength [11 r

(a) [2] The (electric) force between two point charges is (directly) proportional to the product of their charges and inversely proportional to the square of their separation.

(b)(i) [1] The helium nucleus contains two protons, so its charge is +2e.

(b)(ii) [1] $F = Qq/(4\pi\varepsilon_0 r^2) = 2 \times (1.60 \times 10^{-19})^2/[4\pi \times 8.85 \times 10^{-12} \times (170 \times 10^{-12})^2]$
$= 1.6 \times 10^{-8}$ N

(c)(i) [2] The electric force provides the centripetal force: $F = mv^2/r$, so
$v = \sqrt{(Fr/m)} = \sqrt{(1.6 \times 10^{-8} \times 170 \times 10^{-12} / 9.11 \times 10^{-31})}$
$= 1.7 \times 10^6$ m s^{-1}

(c)(ii) [2] $v = 2\pi r/T$, so $T = 2\pi r/v = 2\pi \times 170 \times 10^{-12} / (1.7 \times 10^6)$
$= 6.2 \times 10^{-16}$ s
(Equivalently $F = 4\pi^2 mr/T^2$ gives the same period.)

(d)(i) [2] $E \propto Q/r^2$. The other electron has charge e at a distance of 340 pm, and the nucleus has charge 2e at 170 pm, so

ratio = $[e/(340 \times 10^{-12})^2]/[2e/(170 \times 10^{-12})^2] = (170)^2/(2 \times 340)^2 = 0.13$

(d)(ii) [1] The other electron exerts a small repulsive force, so the resultant inward force is slightly less than 1.6×10^{-8} N — hence the speed of the electron is slightly lower (or its period slightly greater).

9702/44/M/J/25 Q5 — Charged sphere — electric potential, field & capacitance [10 marks]

(a) [2] The work done per unit positive charge in moving a small test charge from infinity to the point.

(b)(i) [2] $E \propto 1/r^2$, so at the surface (radius r) compared with 3r:
$E = 3^2 \times 2.0 \times 10^5 = 1.8 \times 10^6$ N C^{-1}

(b)(ii) [2] $V = Q/4\pi\varepsilon_0 r$ and $E = Q/4\pi\varepsilon_0 r^2$, so $E = V/r$.
$r = V/E = 9.0 \times 10^4/1.8 \times 10^6 = 0.050$ m = 5.0 cm

(b)(iii) [2] $Q = 4\pi\varepsilon_0 Vr = 4\pi \times 8.85 \times 10^{-12} \times 9.0 \times 10^4 \times 0.050$
$= 5.0 \times 10^{-7}$ C

(b)(iv) [2] $C = Q/V = 5.0 \times 10^{-7}/9.0 \times 10^4$
$= 5.6 \times 10^{-12}$ F

9702/41/O/N/25 Q5 — Electric potential — the hydrogen atom dipole [10 marks]

(a) [2] The work done per unit positive charge in moving a small test charge from infinity to the point.

(b)(i) [2] $V = (e/4\pi\varepsilon_0)(1/10 \text{ pm} - 1/110 \text{ pm})$ — the proton contributes $+e/4\pi\varepsilon_0 r$ and the electron $-e/4\pi\varepsilon_0 r$.
$V = (1.60 \times 10^{-19}/4\pi \times 8.85 \times 10^{-12}) \times [(10^{-1} - 110^{-1}) \times 10^{12}] = 130$ V

(b)(ii) [2] $V = (1.60 \times 10^{-19}/4\pi \times 8.85 \times 10^{-12}) \times [(30^{-1} - 90^{-1}) \times 10^{12}]$
$= +32$ V

(b)(iii) [1] A cross at the midpoint between the electron and the proton (x = 60 pm), where the +e and −e potentials cancel.

(b)(iv) [3] A curve from (10, +130) to (110, −130), getting shallower until x = 60 pm, crossing V = 0 at (60, 0), then getting steeper, and passing through (30, +32) and (90, −32).

9702/42/O/N/25 Q6 — Isolated charged sphere — field, potential & discharge [11 marks]

(a) [1] The force per unit positive charge acting on a small test charge placed at the point.

(b)(i) [2] Radial lines, pointing away from the sphere (the sphere is positively charged).

(b)(ii) [2] $C = Q/V$, so $V = Q/C = 83 \times 10^{-12}/(69 \times 10^{-12})$
$= 1.2$ V

(b)(iii) [2] $V = Q/4\pi\varepsilon_0 r$, so $r = Q/4\pi\varepsilon_0 V = 83 \times 10^{-12}/(4\pi \times 8.85 \times 10^{-12} \times 1.2)$
$= 0.62$ m

(b)(iv) [2] $E = Q/4\pi\varepsilon_0 r^2 = 83 \times 10^{-12}/(4\pi \times 8.85 \times 10^{-12} \times 0.62^2)$
$= 1.9$ N C^{-1} (or V m^{-1})

(c) [2] $Q = Q_0 e^{\wedge}(-t/RC)$, so $26 = 83e^{\wedge}(-t/(120 \times 10^6 \times 69 \times 10^{-12}))$
$t = 9.6 \times 10^{-3}$ s

9702/43/O/N/25 Q5 — Electric potential — the hydrogen atom dipole [10 marks]

(a) [2] The work done per unit positive charge in moving a small test charge from infinity to the point.

(b)(i) [2] $V = (e/4\pi\varepsilon_0)(1/10 \text{ pm} - 1/110 \text{ pm})$ — the proton contributes $+e/4\pi\varepsilon_0 r$ and the electron $-e/4\pi\varepsilon_0 r$.
$V = (1.60 \times 10^{-19}/4\pi \times 8.85 \times 10^{-12}) \times [(10^{-1} - 110^{-1}) \times 10^{12}] = 130$ V

(b)(ii) [2] $V = (1.60 \times 10^{-19}/4\pi \times 8.85 \times 10^{-12}) \times [(30^{-1} - 90^{-1}) \times 10^{12}]$
$= +32$ V

(b)(iii) [1] A cross at the midpoint between the electron and the proton (x = 60 pm), where the +e and −e potentials cancel.

(b)(iv) [3] A curve from (10, +130) to (110, −130), getting shallower until x = 60 pm, crossing V = 0 at (60, 0), then getting steeper, and passing through (30, +32) and (90, −32).

9702/44/O/N/25 Q5 — Electric potential & field — proton & charged sphere [9 marks]

(a) [3] The potential is defined as zero at infinity. The proton has a positive charge and therefore repels another positive charge, so work is done to bring a positive charge in from infinity (or work is done by the charges as they separate) — hence the potential near the proton is positive.

(b)(i) [1] An arrow through Q pointing directly away from the centre of the sphere (radially outwards).

(b)(ii) [3] Two lines — one in the top-right quadrant, the other in the bottom-left — passing through (±R, $\pm E_0$) and (±2R, $\pm\frac{1}{4}E_0$), with the curve between −3R and −R increasing in gradient magnitude and curve between R and 3R decreasing in gradient magnitude (nothing drawn between −R and R).

(c) [2] $EP = -Qq/4\pi\varepsilon_0 r = -(1.60 \times 10^{-19})^2/[4\pi \times 8.85 \times 10^{-12} \times 5.3 \times 10^{-11}]$
$= -4.3 \times 10^{-18}$ J

9702/41/M/J/26 Q4 — Electric fields & potentials of a proton–electron pair [9 marks]

(a)(i) [3] At P each charge is at the same distance x, and the field of a point charge is $E = Q/(4\pi\varepsilon_0 r^2)$. The two charges have equal magnitude e, so the two field strengths are equal in magnitude. Both fields point towards the electron and away from the proton — that is, in the same direction — so the fields of the proton and the electron add together.

(a)(ii) [3] $V = Q/(4\pi\varepsilon_0 r)$: the charges have equal magnitude and are equidistant from P, so the two potentials have equal magnitude. The potential due to the proton (positive charge) is positive and the potential due to the electron (negative charge) is negative — the signs are opposite, so the two potentials cancel and the total potential at P is zero.

(b)(i) [2] $F = Q_1Q_2/(4\pi\varepsilon_0 r^2)$, with $Q_1 = Q_2 = e$ and separation $r = 2x$
$F = e^2/(16\pi\varepsilon_0 x^2)$

(b)(ii) [1] $E_P = Qq/(4\pi\varepsilon_0 r) = e^2/(4\pi\varepsilon_0 \times 2x) = e^2/(8\pi\varepsilon_0 x)$, and $F = e^2/(16\pi\varepsilon_0 x^2)$, so $E_P = 2Fx$

9702/42/M/J/26 Q6 — Charged conducting sphere — electric potential & dome capacitance [13 m

(a) [2] The electric potential at a point is the work done per unit positive charge in bringing a small test charge from infinity to the point.

(b)(i) [1] $V_0 = Q/(4\pi\varepsilon_0 R)$, where ε_0 is the permittivity of free space.

(b)(ii) [3] Sketch a curve starting at (R, V_0) that falls with decreasing magnitude of V and of gradient ($V = Q/4\pi\varepsilon_0 x$), passing through $(2R, \frac{1}{2}V_0)$ and $(3R, \frac{1}{3}V_0)$, entirely within the positive V region and approaching the x-axis without crossing it.

(c)(i) [2] $V = Q/(4\pi\varepsilon_0 r) \rightarrow Q = 4\pi\varepsilon_0 rV = 4\pi \times 8.85 \times 10^{-12} \times 0.119 \times (-4.80 \times 10^4)$
$= -6.35 \times 10^{-7}$ C

(c)(ii) [1] The dome stores charge (and hence energy) at a potential — it can be charged and discharged — it acts as a capacitor: its capacitance is the charge stored per unit p.d.

(c)(iii) [2] $C = Q/V = 6.35 \times 10^{-7}/4.80 \times 10^4$
$= 1.32 \times 10^{-11}$ F (13.2 pF)

(c)(iv) [2] $E = \frac{1}{2}QV = \frac{1}{2} \times 6.35 \times 10^{-7} \times 4.80 \times 10^4$
$= 0.0152$ J

9702/43/M/J/26 Q4 — Hydrogen atom — electric field & potential at the midpoint [9 marks]

(a)(i) [3] The magnitudes of the two fields are equal — the proton and the electron have charges of equal magnitude and are at equal distances (x) from P, and $E = Q/4\pi\varepsilon 0r^2$.
The directions are the same: the field due to the proton is away from the proton (to the right at P), and the field due to the electron is towards the electron (also to the right at P).

(a)(ii) [3] The magnitudes of the two potentials are equal, since the charge magnitudes and the distances are equal ($V = Q/4\pi\varepsilon 0r$).
The signs are opposite: the potential due to the proton is positive and the potential due to the electron is negative.

(b)(i) [2] $F = Q1Q2/4\pi\varepsilon 0r^2$ with $Q1 = Q2 = e$ and separation $r = 2x$:
$F = e^2/16\pi\varepsilon 0x^2$

(b)(ii) [1] $EP = e^2/4\pi\varepsilon 0(2x) = e^2/8\pi\varepsilon 0x = 2Fx$ (since $F = e^2/16\pi\varepsilon 0x^2$)

9702/44/M/J/26 Q1 — Electron in circular orbit — electric force, acceleration & period [10 marks]

(a) [2] The electric force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of their separation.

(b)(i) [1] $E_p = Qq/(4\pi\varepsilon_0 r) \rightarrow r = Qq/(4\pi\varepsilon_0 E_p)$
$r = (1.60 \times 10^{-19})^2/(4\pi \times 8.85 \times 10^{-12} \times 1.09 \times 10^{-18}) = 2.56 \times 10^{-38}/1.21 \times 10^{-28}$
$= 2.11 \times 10^{-10}$ m

(b)(ii) [1] $F = Qq/(4\pi\varepsilon_0 r^2) = (1.60 \times 10^{-19})^2/(4\pi \times 8.85 \times 10^{-12} \times (2.11 \times 10^{-10})^2)$
$= 2.56 \times 10^{-38}/4.95 \times 10^{-30} = 5.17 \times 10^{-9}$ N

(b)(iii) [1] $a = F/m = 5.17 \times 10^{-9}/9.11 \times 10^{-31} = 5.68 \times 10^{21}$ m s^{-2}

(b)(iv) [2] $a = r\omega^2$ and $\omega = 2\pi/T$:
$T = 2\pi\sqrt{(r/a)} = 2\pi\sqrt{(2.11 \times 10^{-10}/5.68 \times 10^{21})} = 2\pi \times 1.93 \times 10^{-16}$
$= 1.21 \times 10^{-15}$ s

(c) [3] $E_p = -Qq/(4\pi\varepsilon_0 r)$ decreases (becomes more negative), so the radius of the orbit decreases. The force $F = Qq/(4\pi\varepsilon_0 r^2)$ therefore increases, so the centripetal acceleration $a = F/m$ increases. With $a = r\omega^2$, $\omega = \sqrt{(a/r)}$ increases.

19 Capacitance

9702/41/M/J/24 Q6 — Capacitor discharge — current–time graph, resistance & time constant [10 marks]

(a) [3] The p.d. across the capacitor is proportional to the charge on it, and this p.d. equals the p.d. across the resistor. The current in the resistor is proportional to the p.d. across it, and the current is the rate of decrease of charge on the capacitor. Hence the current is proportional to the charge remaining, so the rate of decrease of current itself decreases as the current falls — giving an exponential decay.

(b)(i) [2] Initial current (at t = 0) I_0 = 0.13 mA (read from Fig. 6.2).

$R = V/I_0 = 12 / (0.13\times10^{-3}) = 9.2\times10^4\ \Omega$

(b)(ii) [3] $I = I_0e^{(-t/\tau)}$; at t = τ the current falls to $I_0e^{-1} = 0.13 \times 0.368 = 0.048$ mA.

From Fig. 6.2, I = 0.048 mA at t = 4.3 s, so τ = 4.3 s.

(c) [2] $\tau = RC$

$C = \tau/R = 4.3 / (9.2\times10^4) = 4.7\times10^{-5}$ F

9702/42/M/J/24 Q6 — Capacitors in series and parallel — stored energy variation [10 marks]

(a) [3] In series the same charge Q is stored on both capacitors (the charging current is the same), and the total p.d. is the sum of the p.d.s: V = VX + VY.

V = Q/C, so Q/CT = Q/CX + Q/CY, hence

1/CT = 1/CX + 1/CY

(b)(i) [2] $E = \frac{1}{2}CV^2$

$V = \sqrt{(2E/C)} = \sqrt{(2 \times 2.5\times10^{-3} / 200\times10^{-6})} = \sqrt{25}$

= 5.0 V (shown)

(b)(ii) [3] Total capacitance = CP + CQ = 200 + 400 = 600 μF

$E = \frac{1}{2}CV^2 = \frac{1}{2} \times 600\times10^{-6} \times 5.0^2 = 7.5\times10^{-3}$ J

= 7.5 mJ

(b)(iii) [2] $E = \frac{1}{2}V^2(CP + CQ)$, so E increases linearly with CQ.

Draw a straight line of positive gradient starting at (0, 2.5 mJ) and passing through (400 μF, 7.5 mJ).

9702/42/O/N/24 Q7 — Charging and sharing capacitors — energy stored [10 marks]

(a) [2] Capacitance is the charge per unit potential difference — the charge on one plate divided by the p.d. between the plates.

(b)(i) [2] A straight line through the origin with positive gradient, ending at the point (V, Q) — charge is proportional to p.d. (C = Q/V).

(b)(ii) [2] $W = \frac{1}{2}QV$. The work done in charging is the area under the charge–p.d. graph, which is a triangle of base V and height Q, so $W = \frac{1}{2} \times Q \times V$.

(c)(i) [3] Charge is conserved and, with the capacitors in parallel, the final p.d. across each is the same: $V' = Q/(C + 3C) = Q/4C = V/4$.

X: final p.d. V/4, final charge Q/4;

Y: final p.d. V/4, final charge 3Q/4.

(c)(ii) [1] Less than — some of the stored energy is dissipated as thermal/internal energy while the charge redistributes.

9702/42/F/M/25 Q5 — Capacitor network — series/parallel & discharge through a resistor [9 mar

(a) [2] For capacitors in series the charge is the same ($Q = Q_1 = Q_2$) and the p.d.s add ($V = V_1 + V_2$). Using V = Q/C:

$Q/C = Q/C_1 + Q/C_2$, which gives $1/C = 1/C_1 + 1/C_2$.

(b) [3] Two equal capacitors C in series give C/2, so the total capacitance is C + C/2 = 3C/2.

The total capacitance equals the gradient of the Q–V graph = $400 \times 10^{-6}/6.0$.

$C = (2 \times 400 \times 10^{-6})/(3 \times 6.0) = 44$ μF

(c)(i) [2] $\tau = RC = 54 \times 10^3 \times (3/2) \times 44 \times 10^{-6}$

= 3.6 s

(c)(ii) [2] $I = I_0e^{(-t/RC)}$, so $0.15 = e^{(-t/3.6)}$

$-t/3.6 = \ln 0.15 = -1.90$

t = 6.8 s

9702/42/M/J/25 Q7 — Capacitor discharge — charge, resistance & time constant [10 marks]

(a) [1] VC = VR — the p.d. across the capacitor equals the p.d. across the resistor (they are connected in parallel).

(b)(i) [2] $C = Q/V = 7.2 \times 10^{-3}/12 = 6.0 \times 10^{-4}$ F
$= 600\ \mu F$

(b)(ii) [2] $R = V/I = 12/(1.5 \times 10^{-3}) = 8000\ \Omega$
$= 8.0\ k\Omega$

(b)(iii) [2] $\tau = RC = 8000 \times 6.0 \times 10^{-4}$
$= 4.8$ s

(c) [3] Fig. 7.2 and Fig. 7.3 show that charge and current are both directly proportional to the p.d. (so $Q \propto I$), and the current is the rate of change of charge. Hence $dQ/dt \propto Q$ — the charge decreases at a rate proportional to itself, giving an exponential variation.

9702/41/O/N/25 Q6 — Capacitors — series/parallel, charge & stored energy [11 marks]

(a) [4] Series: $QS = Q_1 = Q_2$ and $VS = V_1 + V_2$.
Parallel: $QS = Q_1 + Q_2$ and $VS = V_1 = V_2$.

(b)(i) [2] $E = \frac{1}{2}CV^2$, so $V = \sqrt{(2E/C)} = \sqrt{(2 \times 19 \times 10^{-3}/470 \times 10^{-6})}$
$= 9.0$ V

(b)(ii) [2] $Q = CV = 470 \times 10^{-6} \times 9.0$
$= 4.2 \times 10^{-3}$ C

(b)(iii) [3] The total charge is unchanged and the total capacitance is $(470 + 180) \times 10^{-6}$ F.
$E = Q^2/2C = (4.23 \times 10^{-3})^2/(2 \times 650 \times 10^{-6})$
$= 14$ mJ

9702/43/O/N/25 Q6 — Capacitors — series/parallel, charge & stored energy [11 marks]

(a) [4] Series: $QS = Q_1 = Q_2$ and $VS = V_1 + V_2$.
Parallel: $QS = Q_1 + Q_2$ and $VS = V_1 = V_2$.

(b)(i) [2] $E = \frac{1}{2}CV^2$, so $V = \sqrt{(2E/C)} = \sqrt{(2 \times 19 \times 10^{-3}/470 \times 10^{-6})}$
$= 9.0$ V

(b)(ii) [2] $Q = CV = 470 \times 10^{-6} \times 9.0$
$= 4.2 \times 10^{-3}$ C

(b)(iii) [3] The total charge is unchanged and the total capacitance is $(470 + 180) \times 10^{-6}$ F.
$E = Q^2/2C = (4.23 \times 10^{-3})^2/(2 \times 650 \times 10^{-6})$
$= 14$ mJ

9702/41/M/J/26 Q5 — Capacitance — parallel derivation & series stored energy [8 marks]

(a) [2] The capacitance is the charge stored per unit potential difference — the charge on one plate divided by the p.d. between the plates.

(b) [3] In parallel, the p.d. across each capacitor is the same: $V = V_X = V_Y$ (= E).
The total charge is the sum of the charges on the two capacitors: $Q_T = Q_X + Q_Y$.
Using $Q = CV$: $C_T V = C_X V + C_Y V$, hence
$C_T = C_X + C_Y$

(c) [3] For capacitors in series: $1/C_T = 1/C + 1/(2C)$, so $C_T = 2C/3$.
$W = \frac{1}{2}C_T V^2 = \frac{1}{2} \times (2C/3) \times 6.00^2 = 1.30 \times 10^{-3}$
$C = (3 \times 1.30 \times 10^{-3})/6.00^2 = 3.90 \times 10^{-3}/36.0 = 1.08 \times 10^{-4}$ F
$= 108\ \mu F$

9702/43/M/J/26 Q5 — Capacitors in series & parallel — combined capacitance & energy [8 marks

(a) [2] Capacitance is the charge stored per unit potential difference ($C = Q/V$), where the charge is the charge on one plate and the potential difference is the p.d. between the plates.

(b) [3] In parallel, the p.d. across each capacitor is the same: $VX = VY = E$.
Total charge: $QT = QX + QY$, and $Q = CV$:
$CT\ V = CX\ V + CY\ V$
$CT = CX + CY$

(c) [3] Series combination: $1/CT = 1/C + 1/2C \rightarrow CT = 2C/3$
$E = \frac{1}{2}CT\ V^2 = \frac{1}{2} \times (2C/3) \times 6.00^2 = 12C$
$12C = 1.30 \times 10^{-3} \rightarrow C = 1.08 \times 10^{-4}$ F

C = 108 μF

9702/44/M/J/26 Q6 — Capacitor switching circuit — time constant, discharge & minimum p.d. [9 marks]

(a)(i) [2] RC = 2.4 s → R = 2.4/(47×10⁻⁶) = 5.1×10⁴ Ω
= 51 kΩ

(a)(ii) [2] The switch is at Y for 2.5 s in each 5.0 s cycle, so the discharge time is 2.5 s:
V = V₀e^(−t/RC) = 1.50e^(−2.5/2.4) = 1.50×0.353
= 0.53 V

(b) [3] While the switch is at X (t = 0–2.5 s and 5.0–7.5 s) the capacitor re-charges: draw horizontal lines at V_C = 1.50 V. While it is at Y (t = 2.5–5.0 s and 7.5–10.0 s) it discharges: draw smooth exponential decay curves of continuously decreasing negative gradient, falling to a minimum of V_C = 0.53 V.

(c) [2] Doubling the frequency halves the discharge time (2.5 s → 1.25 s) before re-charging, so the capacitor loses less charge and its p.d. does not fall as far — the minimum p.d. is greater than 0.53 V.

20 Magnetic fields

9702/42/F/M/24 Q5 — Alpha particle in a magnetic field — circular motion & velocity selection [1 marks]

(a) [2] Any two of: displacement; velocity; momentum; (centripetal) acceleration; (centripetal) force. Their magnitudes stay constant but their directions vary continuously during the motion (the speed and angular speed are constant).

(b)(i) [3] The magnetic force provides the centripetal force: $Bqv = mv^2/r$. With $v = 2\pi r/T$:
$Bq(2\pi r/T) = m(2\pi r/T)^2/r$
$Bq(2\pi r/T) = m(4\pi^2 r/T^2) \rightarrow B = 2\pi m/(qT)$

(b)(ii) [2] For an alpha particle $m = 4u = 4\times1.66\times10^{-27}$ kg and $q = 2e = 2\times1.6\times10^{-19}$ C:
$B = 2\pi\times4\times1.66\times10^{-27}/(2\times1.6\times10^{-19}\times2.5\times10^{-6}) = 0.052$ T

(b)(iii) [1] The periods are the same — $T = 2\pi m/(Bq)$ is independent of the radius r (B, q and m are unchanged).

(b)(iv) [2] For constant velocity the electric force balances the magnetic force: $qE = Bqv \rightarrow E = Bv$
$E = 0.052\times1.1\times10^6 = 5.7\times10^4$ N C^{-1} (equivalent to V m^{-1})

9702/42/F/M/24 Q6 — Coil in a solenoid — flux linkage, induced e.m.f. & magnetic forces [11 ma

(a)(i) [1] A non-zero horizontal straight line from X to Y — the flux linkage is constant throughout the solenoid.

(a)(ii) [2] Inside the solenoid the magnetic field is uniform, so B is constant (and the area A and number of turns N are constant). Hence the flux linkage $\Phi = BAN$ is constant, giving the horizontal line.

(a)(iii) [2] $\Phi = BAN$, with $A = 0.71$ cm$^2 = 7.1\times10^{-5}$ m^2 and $B = 0.080$ T:
$\Phi = 0.080\times7.1\times10^{-5}\times64 = 3.6\times10^{-4}$ Wb turns

(a)(iv) [3] $E = -d\Phi/dt$. Sketch: a horizontal line at E = 0 from time 0 to time t (B constant); a horizontal line at a positive value of E from t to 2t (B increasing at a steady rate); a horizontal line at a negative value of E from 2t to 4t (B decreasing). Lines of equal magnitude for the two non-zero sections are acceptable.

(b) [3] When the supply is switched on a current flows through every turn of the spring, producing a magnetic field around each turn. The fields of adjacent turns exert forces on each other — turns carrying parallel currents in the same direction attract — so the turns are pulled together and the distance between them decreases (the spring shortens).

9702/42/M/J/24 Q7 — Coil in varying magnetic field — flux linkage & induced e.m.f. [14 marks]

(a) [2] The induced e.m.f. is (directly) proportional to the rate of change of magnetic flux linkage.

(b)(i) [2] Maximum B read from Fig. 7.2 = 7.2 mT
$\Phi = BA = 7.2\times10^{-3} \times 3.2\times10^{-4}$
$= 2.3\times10^{-6}$ Wb

(b)(ii) [3] The e.m.f. is proportional to the gradient of the B–t graph. A tangent drawn at the steepest point of Fig. 7.2 gives $dB/dt \approx 7.5$ mT ms$^{-1} = 7.5$ T s^{-1}.
Maximum rate of change of flux linkage $= N \times A \times (dB/dt) = 340 \times 3.2\times10^{-4} \times 7.5$
$= 0.82$ Wb s^{-1}

(b)(iii) [1] By Faraday's law the induced e.m.f. equals the rate of change of flux linkage, so
$V0 = 0.82$ V

(b)(iv) [3] Sketch a sinusoidal curve of period 2.0 ms from t = 0 to t = 6.0 ms:
all peaks at +V0 (t ≈ 0.5, 2.5, 4.5 ms) and all troughs at −V0 (t ≈ 1.5, 3.5, 5.5 ms)
V = 0 at (and only at) t = 0, 1.0, 2.0, 3.0, 4.0, 5.0 and 6.0 ms.

(b)(v) [3] A = amplitude of the induced e.m.f. = V0
$A = 0.82$ V
B = angular frequency $= 2\pi/T = 2\pi/(2.0\times10^{-3})$
$B = 3.1\times10^3$ rad s^{-1}

9702/41/O/N/24 Q7 — Two parallel wires — magnetic field & force between currents [11 marks]

(a) [2] The magnetic flux density is the force acting per unit current per unit length on a current-carrying conductor placed at right angles to the magnetic field: $B = F/IL$.

(b) [3] Concentric circles around the wire (at least two circles), with the spacing between the circles

increasing as the distance from the wire increases (at least four circles). Arrows on the field lines show the field direction is clockwise (right-hand grip rule for a current into the page).

(c)(i) [2] Each wire is in the magnetic field produced by the current in the other wire. The current in each wire is perpendicular to the magnetic field due to the other wire, so a magnetic force (F = BIL) acts on each wire.

(c)(ii) [1] An arrow starting from wire X and pointing towards wire Y, labelled F — parallel currents in the same direction attract, so the force on X is towards Y.

(c)(iii) [2] The magnetic force on Y has the same magnitude as the force on X (the two forces are a Newton' third law pair), but it acts in the opposite direction.

(c)(iv) [1] There is no change in the direction of the force on X — both the current in X and the magnetic field due to Y are reversed, so the direction of the force (attraction towards Y) is unchanged.

9702/42/O/N/24 Q1 — Rotating wheel in magnetic field — flux & induced e.m.f. [13 marks]

(a)(i) [2] $v = r\omega$
$= 0.85 \times 140$
$= 120\ \text{m s}^{-1}$

(a)(ii) [2] $a = r\omega^2 = 0.85 \times 140^2$
$= 1.7 \times 10^4\ \text{m s}^{-2}$

(b)(i) [2] The direction of the induced e.m.f. is such as to (produce effects that) oppose the change that caused it.

(b)(ii) [1] $T = 2\pi/\omega = 2\pi/140 = 0.045\ \text{s} = 45\ \text{ms}$

(b)(iii) [3] In one revolution the spoke sweeps out the area of the disc, so $\Phi = BA = B\pi r^2$
$= 0.18 \times \pi \times 0.85^2$
$= 0.41\ \text{Wb}$

(b)(iv) [2] $E = \Phi/t = 0.41/0.045$
$= 9.1\ \text{V}$

(b)(v) [1] The force on the spoke must oppose the rotation, i.e. act anticlockwise, so by Fleming's left-hand rule the induced current flows from A to X; hence X is at the higher potential.

9702/42/F/M/25 Q6 — Velocity selector — magnetic field, forces & ion paths [11 marks]

(a) [2] For undeviated motion the forces balance: FB = FE, so Bqu = qE. With E = V/d this gives Bqu = qV/d, so u = V/Bd.

(b) [3] $E_k = \tfrac{1}{2}mu^2$, so $u = \sqrt{(2E_k/m)} = \sqrt{(2 \times 4.1 \times 10^{-17}/3.2 \times 10^{-27})} = 1.6 \times 10^5\ \text{m s}^{-1}$.
$B = V/(ud) = 980/(3.6 \times 10^{-2} \times 1.6 \times 10^5)$
$= 0.17\ \text{T}$

(c) [1] The expression u = V/Bd is independent of mass and charge, so the alpha particle also passes through undeviated at the same speed.

(d) [3] There is no resultant force, so the magnetic force is upwards (equal and opposite to the downward electric force). The ions are positive, so the conventional current is from left to right. Using Fleming's left-hand rule, the magnetic field is into the page.

(e) [2] A curved path inside the plates with a consistent direction of curvature and no discontinuity at entry or in curvature, deflecting upwards (the magnetic force now exceeds the electric force, so the ions are deflected towards the top plate).

9702/42/F/M/25 Q7 — Rod in a magnetic field — Faraday's law & induced e.m.f. [9 marks]

(a) [2] The induced e.m.f. is (directly) proportional to the rate of change of (magnetic) flux (linkage).

(b)(i) [2] The rod accelerates uniformly, so its velocity is proportional to time. Fig. 7.2 shows the e.m.f. is proportional to time, so E is proportional to v.

(b)(ii) [3] With v = at, the distance moved in time Δt is atΔt, so the area swept out is L × atΔt. The flux cut is $\Delta\Phi = B \times (\text{area}) = BLat\Delta t$, and $E = \Delta\Phi/\Delta t = BLat$.

(b)(iii) [2] From Fig. 7.2, $E = 0.30 \times 10^{-3}$ V at t = 2.0 s, so
$B = E/(Lat) = (0.30 \times 10^{-3})/(0.45 \times 7.8 \times 2.0)$
$= 4.3 \times 10^{-5}\ \text{T}$

9702/41/M/J/25 Q7 — Magnetic fields — flux density, force & velocity selector [10 marks]

(a) [2] Magnetic flux density is the force per unit length per unit current acting on a (straight) conductor placed perpendicular to the field.

(b)(i) [1] F = BQv (with the velocity perpendicular to the magnetic field).

(b)(ii) [1] An arrow at Y pointing vertically upwards (from Fleming's left-hand rule for a positive charge).

(b)(iii) [1] A circular path curving upwards through the field (the force is always perpendicular to the velocity, so it acts as a centripetal force).

(c)(i) [3] Apply an electric field vertically downwards so that the electric force on the particle is in the opposite direction to the magnetic force. The particle then passes straight through to Z when the magnitudes of the electric and magnetic forces are equal.

(c)(ii) [2] EQ = BQv, so v = E/B.

9702/42/M/J/25 Q2 — Gravitational & magnetic fields of the Earth [12 marks]

(a)(i) [2] A gravitational field line shows the direction of the (gravitational) force that would act on a small test mass placed at that point.

(a)(ii) [2] At least four radial lines from the Earth's surface, evenly spaced around it, with arrows pointing towards the centre of the Earth.

(b)(i) [1] The top pole is labelled S and the bottom pole is labelled N (the geographic north pole is near the Earth's magnetic south pole).

(b)(ii) [2] At the poles: at least two field lines on either side of each pole, clustered close together, leaving the surface approximately perpendicular to it and curving away from the pole axis. Above the equator: at least one field line on each side looping from one hemisphere to the other, passing above the surface near the equator approximately parallel to the surface.

(c)(i) [2] The gravitational field lines are evenly spaced around the surface and all point perpendicular to the surface (towards the centre), so the field has the same magnitude and direction at every point on the surface — it does not vary around the surface.

(c)(ii) [3] The magnetic field is strongest at the poles and weakest near the Equator. It is perpendicular to the surface at the poles and parallel to the surface near the Equator, with the angle to the surface increasing from the Equator towards the poles.

9702/43/M/J/25 Q7 — Magnetic fields — flux density, force & velocity selector [10 marks]

(a) [2] Magnetic flux density is the force per unit length per unit current acting on a (straight) conductor placed perpendicular to the field.

(b)(i) [1] F = BQv (with the velocity perpendicular to the magnetic field).

(b)(ii) [1] An arrow at Y pointing vertically upwards (from Fleming's left-hand rule for a positive charge).

(b)(iii) [1] A circular path curving upwards through the field (the force is always perpendicular to the velocity, so it acts as a centripetal force).

(c)(i) [3] Apply an electric field vertically downwards so that the electric force on the particle is in the opposite direction to the magnetic force. The particle then passes straight through to Z when the magnitudes of the electric and magnetic forces are equal.

(c)(ii) [2] EQ = BQv, so v = E/B.

9702/44/M/J/25 Q6 — Coil in a magnetic field — force & torque [10 marks]

(a)(i) [3] $F = BIL \times N = 5.2 \times 10^{-3} \times 1.2 \times 0.054 \times 190$
$= 0.064$ N

(a)(ii) [2] torque = force × perpendicular distance between the forces on QR and PS
$= 0.064 \times 0.025 \cos\theta = 1.6 \times 10^{-3} \cos\theta$ N m

(a)(iii) [3] One complete cycle of a sinusoidal curve between 0° and 360°, with the τ axis labelled to show maximum and minimum torque at $\pm 1.6 \times 10^{-3}$ N m: maximum at 0° and 360°, minimum at 180° torque zero at 90° and 270°.

(b) [2] The ferrous core increases the magnetic flux density, so the amplitude of the torque increases.

9702/41/O/N/25 Q7 — Faraday's law — induced e.m.f. across an aircraft's wings [11 marks]

(a) [2] The induced e.m.f. is (directly) proportional to the rate of change of magnetic flux (linkage).

(b)(i) [2] e.m.f. = $\Delta\Phi/\Delta t$, so flux = e.m.f. × time = 0.54 × 15
= 8.1 Wb

(b)(ii) [2] $\Phi = BA$, so area = $\Phi/B = 8.1/(38 \times 10^{-6})$
$= 2.1 \times 10^{5}$ m^2

(b)(iii) [2] area = speed × time × width, so v = area/(time × width) = $2.1 \times 10^5/(15 \times 68)$
$= 210$ m s^{-1}

(b)(iv) [3] By Lenz's law the induced current produces an opposing force which must act backwards. From

Fleming's left-hand rule the current in the wings is therefore from Q to P. Inside an e.m.f. source the current flows from − to +, so P is at the higher potential.

9702/43/O/N/25 Q7 — Faraday's law — induced e.m.f. across an aircraft's wings [11 marks]

(a) [2] The induced e.m.f. is (directly) proportional to the rate of change of magnetic flux (linkage).

(b)(i) [2] e.m.f. = $\Delta\Phi/\Delta t$, so flux = e.m.f. × time = 0.54 × 15
= 8.1 Wb

(b)(ii) [2] $\Phi = BA$, so area = $\Phi/B = 8.1/(38 \times 10^{-6})$
$= 2.1 \times 10^5$ m²

(b)(iii) [2] area = speed × time × width, so v = area/(time × width) = $2.1 \times 10^5/(15 \times 68)$
= 210 m s^{-1}

(b)(iv) [3] By Lenz's law the induced current produces an opposing force which must act backwards. From Fleming's left-hand rule the current in the wings is therefore from Q to P. Inside an e.m.f. source the current flows from − to +, so P is at the higher potential.

9702/44/O/N/25 Q7 — Lenz's law — induced e.m.f. across a helicopter rotor [9 marks]

(a) [2] The induced e.m.f. acts in a direction such as to oppose the change that caused it.

(b)(i) [3] $\Phi = BA = 0.047 \times 10^{-3} \times \pi \times 12^2$
= 0.021 Wb

(b)(ii) [2] e.m.f. = flux cut/time = 0.021/(1/85)
= 1.8 V

(b)(iii) [2] The force due to the induced current produces an anticlockwise moment (opposing the rotation). From the left-hand rule the current is from O to X, so X is at the higher potential.

9702/41/M/J/26 Q6 — Charged particles in a magnetic field — mass spectrometer [12 marks]

(a) [2] Magnetic flux density is the force per unit current per unit length acting on a current-carrying wire placed at right angles to the magnetic field ($B = F/IL$ with the wire perpendicular to the field).

(b)(i) [2] At X the (magnetic) force on the particles is to the left, curving the beam towards the detector at Y. Applying Fleming's left-hand rule with the field out of the plane of the page and the effective current in the direction of movement of the particles gives a force in this direction only for a positive charge — hence the particles are positively charged.

(b)(ii) [2] The magnetic force is always perpendicular to the direction of motion of the particles, so it does no work and changes only the direction, not the speed. A force of constant magnitude alway perpendicular to the velocity provides the centripetal acceleration, so the particles follow a circular path.

(c)(i) [2] XY is the diameter of the circular path, so r = ½ × 8.2×10^{-2} = 0.041 m.
$a = v^2/r = (7.3 \times 10^4)^2/0.041 = 5.33 \times 10^9/0.041$
$= 1.3 \times 10^{11}$ m s^{-2}

(c)(ii) [2] $Bqv = ma \rightarrow B = ma/(qv)$
$= (2.0 \times 10^{-26} \times 1.3 \times 10^{11})/(3.2 \times 10^{-19} \times 7.3 \times 10^4) = 2.6 \times 10^{-15}/2.34 \times 10^{-14}$
= 0.11 T

(d) [2] $r = mv/(Bq)$. With the mass quartered and the charge halved, $r' = (m/4)v/(B \times q/2) = r/2$ — the radius is halved. Draw the beam entering the field at X along the same line as before, deflecting in the same direction, but following a circular path of half the radius.

9702/42/M/J/26 Q7 — Hall probe in a magnetic field — Hall voltage & charge carriers [8 marks]

(a)(i) [1] The Hall voltage is induced between faces PSZW and QRYX.

(a)(ii) [3] n = number density of charge carriers (number per unit volume);
q = charge on each charge carrier;
t = thickness of the slice (the distance between faces PSZW and QRYX).

(b)(i) [3] Any three of: the charge carriers experience a force due to the magnetic field; the force on the carriers is perpendicular to both I and B; the carriers deflect towards one face (accumulating there, leaving the opposite face with the opposite charge); the charge separation between the faces produces a p.d. (the Hall voltage) between them; a steady state is reached when the magnetic force is balanced by the force due to the resulting electric field.

(b)(ii) [1] The (negative) electrons are deflected towards face QRYX, leaving face PSZW positively charged - face PSZW is at the higher potential.

9702/43/M/J/26 Q6 — Mass spectrometer — circular path & magnetic flux density [12 marks]

(a) [2] Magnetic flux density is the force per unit current per unit length on a wire placed at right angles to the magnetic field (B = F/IL).

(b)(i) [2] At X, the magnetic force on the particles is to the left (the path curves towards Y). Applying Fleming's left-hand rule with the force to the left and the field out of the page shows that the (effective) current is in the direction of movement of the particles — so the charge is positive.

(b)(ii) [2] The magnetic force is always perpendicular to the direction of motion of the particles. It changes only the direction of the motion, not the speed — it acts as a centripetal force, so the path is circular (at constant speed).

(c)(i) [2] The path is a semicircle, so r = ½XY = ½ × 8.2 = 4.1 cm = 0.041 m.
$a = v^2/r = (7.3\times10^4)^2/0.041 = 5.33\times10^9/0.041 = 1.3\times10^{11}$ m s^{-2} (shown)

(c)(ii) [2] BQv = ma
$B = ma/Qv = (2.0\times10^{-26} \times 1.3\times10^{11})/(3.2\times10^{-19} \times 7.3\times10^4) = 2.6\times10^{-15}/2.34\times10^{-14}$
B = 0.11 T

(d) [2] Radius of the circular path: r = mv/BQ. With the mass quartered and the charge halved, r′ = r × ¼ ÷ ½ = r/2.

Draw the beam entering the field at X along the same line and deflecting in the same direction, following a semicircular path of half the radius of the original beam.

9702/44/M/J/26 Q7 — Solenoid magnetic field — flux linkage & induced e.m.f. [14 marks]

(a) [1] An alternating current is a current whose direction regularly (periodically) reverses.

(b)(i) [3] Draw at least five straight, parallel, horizontal field lines inside the solenoid; outside, draw the bar-magnet pattern of curved lines looping from one end to the other (at least one complete loop above and one below the central axis, plus one horizontal line outside each pole). The inside and outside lines must be continuous with each other, with arrows showing the field direction right to left inside the solenoid.

(b)(ii) [1] Mark N at the left-hand end and S at the right-hand end of the solenoid.

(c)(i) [1] B = 2.0 cos(200πt), so ω = 200π = 2π/T:
T = 2π/200π = 0.010 s

(c)(ii) [2] Draw two complete cycles of a cosine curve with period 0.010 s and amplitude 2.0 mT: B = +2.0 m at t = 0, zero at t = 0.0025, 0.0075, 0.0125 and 0.0175 s, with minima of −2.0 mT at t = 0.005 s and t = 0.015 s.

(d)(i) [3] $\Phi = BAN = 2.0\times10^{-3}\times\pi\times(2.8\times10^{-3})^2\times45$
$= 2.0\times10^{-3}\times2.46\times10^{-5}\times45$
$= 2.2\times10^{-6}$ Wb

(d)(ii) [3] Draw a tangent to the B–t curve at its steepest part and measure its gradient (the maximum rate of change of B). Multiply this steepest gradient by the area of the coil and the number of turns to obtain the maximum rate of change of flux linkage. By Faraday's law the maximum induced e.m.f. equals this maximum rate of cutting flux.

21 Alternating currents

9702/42/F/M/24 Q4 — Capacitor network — bridge rectifier, smoothing & mean power [10 marks]

(a) [2] 10 μF and 20 μF in parallel give 30 μF; this is in series with the 45 μF capacitor:
1/C = 1/45 + 1/30 → C = 18 μF

(b) [2] $\Delta E = \frac{1}{2}C(V_2^2 - V_1^2) = \frac{1}{2}\times45\times10^{-6}\times(9.6^2 - 8.0^2)$
$= \frac{1}{2}\times45\times10^{-6}\times28.16 = 6.3\times10^{-4}$ J

(c)(i) [1] Add a capacitor in parallel with the load resistor, i.e. connected directly across the two output terminals of the bridge rectifier, with wires completing the circuit.

(c)(ii) [3] From Fig. 4.3: V = 4.0 V at t = 5 ms and V = 3.2 V at t = 13 ms (a drop over 8 ms). Using V = V_0e^(−t/RC):
3.2 = 4.0e^(−0.008/RC)
RC = −0.008/ln(3.2/4.0) = 36 ms

(d)(i) [1] Mean power = half the maximum power for a sinusoidal a.c.:
mean power = 16/2 = 8.0 W

(d)(ii) [1] Half-wave rectification supplies power for only half of each cycle, so the mean power is half of the full-wave value:
mean power = 8.0/2 = 4.0 W

9702/41/M/J/24 Q7 — Full-wave rectification — power dissipation in a load resistor [10 marks]

(a) [2] The circuit rectifies the alternating input voltage — a bridge rectifier of four diodes that produces a full-wave rectified (unidirectional) output voltage across R.

(b)(i) [2] Peak output voltage V_0 = 9.0 V (from Fig. 7.2).
$P_{max} = V_0^2/R = 9.0^2 / 370 = 81 / 370 = 0.22$ W

(b)(ii) [3] A sinusoidal curve that sits on (or above) the time axis — minima of zero power at t = 0, 0.02, 0.04, 0.06 and 0.08 s, and maxima of 0.22 W at t = 0.01, 0.03, 0.05 and 0.07 s (period 0.02 s, i.e. frequency 50 Hz).

(b)(iii) [1] Mean power = peak power / 2 = 0.22 / 2 = 0.11 W

(c) [2] The power–time graph is identical to that in (b)(ii) — the power depends on the square of the voltage, so the negative half-cycles give the same power pulses as the rectified positive ones. Hence the mean power dissipated is the same (0.11 W).

9702/41/O/N/24 Q6 — Half-wave rectification & capacitor smoothing — capacitance & resistance marks]

(a)(i) [1] Rectification is the conversion of an alternating voltage (a.c.) into a direct voltage (d.c.) — a unidirectional output voltage.

(a)(ii) [2] Half-wave rectification removes the voltage in one direction (the output is zero during alternate half-cycles), whereas full-wave rectification reverses the voltage in one direction so that the output voltage is in the same direction throughout each cycle.

(b)(i) [2] In Fig. 6.1, one of the gaps is connected by a single diode (drawn with the correct circuit symbol — a triangle with a bar) and the other gap is connected directly by a wire. The diode conducts in only one half of each cycle, so a p.d. appears across R in that half-cycle only.

(b)(ii) [1] Smoothing — the capacitor charges when VOUT rises and discharges through R when VOUT falls, reducing the variation (ripple) of the output voltage.

(c)(i) [2] $E = \frac{1}{2}CV^2 \rightarrow C = 2E/V^2$
$C = 2 \times 0.041/12^2 = 5.7\times10^{-4}$ F = 570 μF

(c)(ii) [3] From Fig. 6.3, VOUT decays from 12 V to 8.0 V in 0.010 s (during the non-conducting half-cycle).
V = V_0e^(−t/RC): 8.0 = 12.0e^(−0.010/RC)
ln(8.0/12.0) = −0.010/RC, so RC = 0.010/ln(1.5) = 0.0247 s
$R = RC/C = 0.0247/(5.7\times10^{-4}) = 43\ \Omega$

9702/42/O/N/24 Q8 — Sinusoidal alternating current — r.m.s. & mean power [9 marks]

(a) [1] Frequency is the number of complete cycles per unit time.

(b)(i) [1] $T = 2\pi/\omega = 2\pi/40\pi = 0.050$ s = 50 ms

(b)(ii) [3] A sinusoidal curve starting at (0, 0) and initially increasing, showing two complete cycles between t = 0 and t = 100 ms (period 50 ms), with all peaks at I = +3.5 A and all troughs at I = −3.5 A.

(b)(iii) [1] $I_{r.m.s.} = I_0/\sqrt{2} = 3.5/\sqrt{2}$
$= 2.5$ A
(c) [3] Peak power $= I_0^2R = 3.5^2 \times 680 = 8330$ W.
Mean power $= I_{r.m.s.}^2R = 2.47^2 \times 680 = 4150$ W $\approx \frac{1}{2} \times 8330$ W, so the mean power is half the peak power.

9702/41/M/J/25 Q6 — Rectification & smoothing — a.c. to d.c. [12 marks]

(a)(i) [1] The conversion of an alternating current (a.c.) into a direct current (d.c.).
(a)(ii) [1] To smooth the rectified output (reduce the ripple).
(b)(i) [2] A = 12 V (the peak value), and $B = 2\pi/T = 2\pi/(20 \times 10^{-3})$
$= 310$ rad s^{-1}
(b)(ii) [1] Full-wave rectification (both halves of each cycle are used).
(b)(iii) [2] Four diodes connected as a bridge rectifier — a diamond/lattice of four diodes (each with the correct circuit symbol), with the a.c. input connected across one diagonal and the output taken across the other.
(b)(iv) [3] During discharge $V = V_0e^{(-t/\tau)}$. From Fig. 6.2 the p.d. falls from 12 V to 8.0 V in 7.3 ms:
$8.0 = 12e^{(-7.3 \times 10^{-3}/\tau)}$
$\tau = -7.3 \times 10^{-3}/\ln(8.0/12) = 0.018$ s
(c) [2] $\tau = RC$, so $R = \tau/C = 0.018/(570 \times 10^{-6})$
$= 32\ \Omega$

9702/42/M/J/25 Q8 — Alternating current — half-wave rectification & power [10 marks]

(a)(i) [1] Half-wave rectification.
(a)(ii) [1] $V_0 = \sqrt{2} \times V_{r.m.s.} = \sqrt{2} \times 6.0$
$= 8.5$ V
(b)(i) [2] $P = V^2/R$, so $P_0 = V_0^2/R = 8.5^2/45$
$= 1.6$ W
(b)(ii) [3] Over two cycles (t = 0 to 2T): two humps of width 0.5T separated by two sections of zero power of width 0.5T, with peaks at P_0, a smooth sinusoidal shape, and the troughs sitting on the t-axis at P = 0.
(b)(iii) [2] Within each hump the mean power is $\frac{1}{2}P_0$ (the average of a $\sin^2$ half-cycle), and only half of each cycle carries any power, so the overall mean power is $\frac{1}{2} \times \frac{1}{2}P_0 = \frac{1}{4}P_0$.
(b)(iv) [1] $\langle P \rangle = V_{r.m.s.}^2/R$, so $V_{r.m.s.} = \sqrt{(\langle P \rangle R)} = \sqrt{((P_0/4) \times 45)} = \sqrt{(1.6 \times 45/4)}$
$= 4.2$ V

9702/43/M/J/25 Q6 — Rectification & smoothing — a.c. to d.c. [12 marks]

(a)(i) [1] The conversion of an alternating current (a.c.) into a direct current (d.c.).
(a)(ii) [1] To smooth the rectified output (reduce the ripple).
(b)(i) [2] A = 12 V (the peak value), and $B = 2\pi/T = 2\pi/(20 \times 10^{-3})$
$= 310$ rad s^{-1}
(b)(ii) [1] Full-wave rectification (both halves of each cycle are used).
(b)(iii) [2] Four diodes connected as a bridge rectifier — a diamond/lattice of four diodes (each with the correct circuit symbol), with the a.c. input connected across one diagonal and the output taken across the other.
(b)(iv) [3] During discharge $V = V_0e^{(-t/\tau)}$. From Fig. 6.2 the p.d. falls from 12 V to 8.0 V in 7.3 ms:
$8.0 = 12e^{(-7.3 \times 10^{-3}/\tau)}$
$\tau = -7.3 \times 10^{-3}/\ln(8.0/12) = 0.018$ s
(c) [2] $\tau = RC$, so $R = \tau/C = 0.018/(570 \times 10^{-6})$
$= 32\ \Omega$

9702/44/M/J/25 Q8 — Bridge rectifier & smoothing — time constant [10 marks]

(a) [2] Four diodes (one in each gap of the bridge), each with the correct circuit symbol, all connected in the correct direction (pointing left to right) so that A is at a positive potential with respect to B.
(b)(i) [3] Three lines showing exponential decay from the peaks at 0.5T, 1.0T and 1.5T, ending where the decay meets the next rising peak, with rising portions following the dotted peaks, and a minimum V of $(2/3)V_0$.
(b)(ii) [3] From the graph the discharge time (from V_0 to $(2/3)V_0$) is (11/30)T.

(2/3)V_0 = V_0 e^(−(11/30)T/τ)
ln(2/3) = −11T/30τ
τ = 0.90 T

(b)(iii) [2] The time constant increases (τ = RC increases with R), so the capacitor discharges more slowly: the minimum voltage is larger (a smaller difference between the maximum and minimum voltage

9702/42/O/N/25 Q7 — Alternating voltage — period, r.m.s. & rectification [8 marks]

(a)(i) [1] T = 2π/ω = 2π/40π = 0.050 s

(a)(ii) [1] $V_{r.m.s.}$ = 18/√2 = 13 V

(b) [3] A sinusoidal curve of period 50 ms from t = 0 to t = 100 ms, with Vmax at t = 0, 50, 100 ms and −Vmax at t = 25, 75 ms, and the maxima/minima at ±18 V.

(c) [3] Any three of: the rectification is full-wave; the mean power is 14 W; the resistance of R is 12 Ω; the peak current in R is 1.6 A (r.m.s. current 1.1 A); the period of the output is 25 ms (frequency 40 Hz, or angular frequency 250 rad s^{-1}).

9702/44/O/N/25 Q6 — Bridge rectifier & smoothing — a.c. to d.c. [15 marks]

(a)(i) [1] The conversion of an alternating current (a.c.) into a direct current (d.c.).

(a)(ii) [1] Full-wave rectification.

(a)(iii) [2] The three missing diodes with correct symbols: the lower-left diode pointing towards the right, and the upper-left and lower-right diodes both pointing towards the left.

(b)(i) [2] ω = 18 rad s^{-1}, so T = 2π/ω = 2π/18 = 0.35 s

(b)(ii) [1] $V_{r.m.s.}$ = 34/√2 = 24 V

(b)(iii) [3] Two sinusoidal 'humps' (full-wave rectified), all the same sign, with peak VOUT = 34 V and VOUT = 0 at t = 0, 0.175 s and 0.35 s (non-zero in between).

(c)(i) [1] A capacitor connected in parallel with the resistor.

(c)(ii) [2] τ = RC = 56 × 10^3 × 12 × 10^{-6} = 0.67 s

(c)(iii) [2] Vmin = 34e^(−0.14/0.67) = 28 V

9702/41/M/J/26 Q7 — Half-wave rectification & r.m.s. values [8 marks]

(a)(i) [2] Half-wave rectification allows the current to pass (unhindered) in one direction and blocks the current (completely) in the other direction, so the output contains only one polarity of the alternating input.

(a)(ii) [2] Complete the circuit by inserting a diode in one of the two gaps and a direct wire connection across the other gap. The diode must be drawn with the correct diode symbol (triangle with bar) and orientated so that it conducts for one polarity of the input.

(b) [4] Complete table:

V_1 (sinusoidal): unrectified = V_0/√2, half-wave rectified = V_0/2

V_2 (square wave): unrectified = V_0, half-wave rectified = V_0/√2

9702/42/M/J/26 Q8 — Rectification & smoothing of a sinusoidal input voltage [7 marks]

(a)(i) [1] Diode (single diode producing half-wave rectification).

(a)(ii) [1] Capacitor (connected in parallel across the output).

(b)(i) [1] peak value of VIN = 3.6 V

(b)(ii) [1] period of VIN = 0.040 s

(c) [3] Any three of: the rectification is half-wave (VOUT is zero during alternate half-cycles); frequency of VIN = 1/0.040 = 25 Hz (angular frequency 160 rad s^{-1}); r.m.s. input voltage = 3.6/√2 = 2.5 V; time constant of the smoothing circuit = 0.058 s; ripple (variation in VOUT) = 1.6 V (44% of the peak).

9702/43/M/J/26 Q7 — Half-wave rectification — r.m.s. values of two waveforms [8 marks]

(a)(i) [2] Half-wave rectification allows the current (unhindered) in one direction and blocks the current completely in the other direction, so that the output current is always in the same direction.

(a)(ii) [2] Complete the circuit by drawing a diode (with the correct diode symbol) in one of the two gaps, connected so that it conducts in the required direction, and a direct connection across the other gap.

(b) [4] Complete the table as follows:

V1 (sinusoidal), unrectified: V0/√2

V2 (square wave), unrectified: V0

V1, half-wave rectified: V0/2
V2, half-wave rectified: V0/√2
(Half-wave rectification halves the mean-square value of the sinusoid — Vrms = √(V0²/4) = V0/2 — while the rectified square wave is V0 for half the period and 0 for half — Vrms = √(V0²/2) = V0/√2.)

22 Quantum physics

9702/42/F/M/24 Q7 — Photon momentum & the photoelectric effect — threshold wavelength [9 m

(a) [2] p = E/c = $3.11\times10^{-19}/3.00\times10^{8} = 1.04\times10^{-27}$ N s

(b)(i) [2] E(photon) = hc/λ = $6.63\times10^{-34}\times3.00\times10^{8}/(640\times10^{-9}) = 3.11\times10^{-19}$ J

N = Pt/E = $0.350\times1.0/3.11\times10^{-19} = 1.1\times10^{18}$ photons

(b)(ii) [2] Momentum of one photon p = E/c. In time t the laser emits N = Pt/E photons, so the total momentum delivered to the surface is NE/c = Pt/c. Force = rate of change of momentum = (Pt/c)/t so F = P/c.

(c)(i) [1] The threshold wavelength is the maximum wavelength of electromagnetic radiation that causes emission of electrons from the surface of the metal (the photon energy then equals the work function energy).

(c)(ii) [2] The largest threshold wavelength corresponds to the smallest work function: potassium, Φ = 2.26 eV = $2.26\times1.6\times10^{-19} = 3.62\times10^{-19}$ J

λ_0 = hc/Φ = $6.63\times10^{-34}\times3.00\times10^{8}/(3.62\times10^{-19}) = 5.50\times10^{-7}$ m

9702/41/M/J/24 Q8 — X-ray tube — photon energy, momentum & wavelength [13 marks]

(a) [2] A photon is a quantum (packet) of electromagnetic radiation, carrying energy E = hf.

(b)(i) [1] Electrons (accelerated from the filament towards the metal target).

(b)(ii) [1] Terminal X is labelled − and terminal Y is labelled + (electrons, being negative, are accelerated from the negative filament towards the positive target).

(c)(i) [1] Maximum photon energy = eV = 32 keV = 0.032 MeV

(c)(ii) [2] p = E/c = $(0.032 \times 1.60\times10^{-13}) / (3.00\times10^{8}) = 5.12\times10^{-15} / 3.00\times10^{8} = 1.7\times10^{-23}$ N s

(c)(iii) [3] E = hf and c = fλ → λ = hc/E

= $(6.63\times10^{-34} \times 3.00\times10^{8}) / (0.032 \times 1.60\times10^{-13}) = 1.99\times10^{-25} / 5.12\times10^{-15} = 3.9\times10^{-11}$ m

(d) [3] X-rays are attenuated differently by different tissues — bone and soft tissue have very different linear attenuation (absorption) coefficients, so the X-rays transmitted through bone and through soft tissue have very different intensities. The large difference in transmitted intensity produces the good contrast in the image.

9702/41/O/N/24 Q8 — Photoelectric emission from magnesium — work function & electron speed marks]

(a) [1] The photoelectric effect.

(b)(i) [2] Φ = $hf_0 = 6.63\times10^{-34} \times 8.8\times10^{14} = 5.8\times10^{-19}$ J

(b)(ii) [3] hf = Φ + ½mvmax²

$6.63\times10^{-34} \times 11\times10^{14} = 5.8\times10^{-19} + ½ \times 9.11\times10^{-31} \times v_{max}^2$

$v_{max}^2 = 2 \times (7.29\times10^{-19} - 5.8\times10^{-19})/(9.11\times10^{-31}) = 3.2\times10^{11}$

$v_{max} = 5.7\times10^{5}$ m s⁻¹

(c) [3] EMAX = hf − Φ: EMAX is zero from f = 8.0×10^{14} to 8.8×10^{14} Hz (below the threshold frequency), then rises as a single straight line of positive gradient (gradient = h) from f = 8.8×10^{14} to 11×10^{14} Hz, passing through the point (11, 1.45) — EMAX = 1.45×10^{-19} J at f = 11×10^{14} Hz.

9702/42/O/N/24 Q9 — Electron diffraction — de Broglie wavelength & momentum [9 marks]

(a) [1] Diffraction is characteristic of wave behaviour, so the observation shows that electrons can behave like waves.

(b) [3] qV = ½mv² → v = √(2qV/m)

p = mv = m√(2qV/m)

= √(2qVm)

(c) [2] The electrons gain greater kinetic energy, so they have greater momentum and hence a smaller de Broglie wavelength (λ = h/p). The interference fringes therefore become closer together.

(d)(i) [2] λ = h/p, so p = h/λ = h × (1/λ): a straight line with positive gradient passing through the origin.

(d)(ii) [1] The Planck constant.

9702/42/F/M/25 Q8 — Laser photons — energy, wavelength & radiation force [10 marks]

(a) [2] A photon is a quantum of energy of electromagnetic radiation.

(b)(i) [3] ΔE = hc/λ, so λ = hc/ΔE

= (6.63 × 10^{-34} × 3.00 × 10^{8})/(1.96 × 1.60 × 10^{-19})
= 6.3 × 10^{-7} m

(b)(ii) [1] Number per unit time = power/energy per photon
= (1.0 × 10^{-2})/(1.96 × 1.60 × 10^{-19})
= 3.2 × 10^{16} s^{-1}

(b)(iii) [4] Force = rate of change of momentum, and for a photon p = E/c.
Half the photons are absorbed (momentum change p) and half are reflected (momentum change 2p), so the average momentum change per photon is (p + 2p)/2 = 3p/2.
F = (3/2) × (E/c) × (number per second)
= (3/2) × [(1.96 × 1.60 × 10^{-19})/(3.00 × 10^{8})] × 3.2 × 10^{16}
= 5.0 × 10^{-11} N

9702/41/M/J/25 Q8 — Electron & positron — de Broglie wavelength & annihilation [13 marks]

(a) [1] The wavelength associated with a moving particle.

(b) [2] λ = h/p = h/mv = 6.63 × 10^{-34}/(9.11 × 10^{-31} × 4.9 × 10^{7})
= 1.5 × 10^{-11} m

(c) [2] Similarity: they have the same mass (or the same magnitude of charge, or both are leptons).
Difference: the electron has negative charge while the positron has positive charge (the positron is the antiparticle of the electron).

(d)(i) [1] Annihilation (a particle–antiparticle reaction).

(d)(ii) [2] The electron and positron disappear — their mass is converted into energy, and this mass–energy becomes the energy of the two gamma-ray photons.

(d)(iii) [1] Two photons are produced travelling in opposite directions so that momentum is conserved (the initial total momentum is zero).

(d)(iv) [1] Ek = ½mv² = ½ × 9.11 × 10^{-31} × (4.9 × 10^{7})²
= 1.1 × 10^{-15} J

(d)(v) [3] Each of the two gamma photons carries the rest energy of one particle plus its kinetic energy:
E = mc² + Ek = 9.11 × 10^{-31} × (3.00 × 10^{8})² + 1.1 × 10^{-15} = 8.3 × 10^{-14} J.
Using E = hc/λ:
λ = hc/E = (6.63 × 10^{-34} × 3.00 × 10^{8})/(8.3 × 10^{-14}) = 2.39 × 10^{-12} m

9702/42/M/J/25 Q9 — Photoelectric effect — threshold frequency & work function [8 marks]

(a) [2] The emission of electrons from a metal surface when electromagnetic radiation is incident on it.

(b)(i) [3] VS is the stopping voltage — the current falls to zero when the applied p.d. equals the energy per unit charge of the most energetic electrons. The energy of a photon depends on the frequency (E = hf), so the maximum energy of the emitted electrons depends on the photon energy, and therefore VS depends on the frequency.

(b)(ii) [3] Any three of: threshold frequency = 1.5 × 10^{15} Hz; threshold wavelength = 2.0 × 10^{-7} m; work function = 6.2 eV (9.9 × 10^{-19} J); Planck constant = 6.6 × 10^{-34} J s; number of photons/electrons per second = 1.7 × 10^{16} s^{-1} (in stage 1); power of the incident radiation = 0.028 W.

9702/43/M/J/25 Q8 — Electron & positron — de Broglie wavelength & annihilation [13 marks]

(a) [1] The wavelength associated with a moving particle.

(b) [2] λ = h/p = h/mv = 6.63 × 10^{-34}/(9.11 × 10^{-31} × 4.9 × 10^{7})
= 1.5 × 10^{-11} m

(c) [2] Similarity: they have the same mass (or the same magnitude of charge, or both are leptons).
Difference: the electron has negative charge while the positron has positive charge (the positron is the antiparticle of the electron).

(d)(i) [1] Annihilation (a particle–antiparticle reaction).

(d)(ii) [2] The electron and positron disappear — their mass is converted into energy, and this mass–energy becomes the energy of the two gamma-ray photons.

(d)(iii) [1] Two photons are produced travelling in opposite directions so that momentum is conserved (the initial total momentum is zero).

(d)(iv) [1] Ek = ½mv² = ½ × 9.11 × 10^{-31} × (4.9 × 10^{7})²
= 1.1 × 10^{-15} J

(d)(v) [3] Each of the two gamma photons carries the rest energy of one particle plus its kinetic energy:

$E = mc^2 + Ek = 9.11 \times 10^{-31} \times (3.00 \times 10^8)^2 + 1.1 \times 10^{-15} = 8.3 \times 10^{-14}$ J.
Using $E = hc/\lambda$:
$\lambda = hc/E = (6.63 \times 10^{-34} \times 3.00 \times 10^8)/(8.3 \times 10^{-14}) = 2.39 \times 10^{-12}$ m

9702/44/M/J/25 Q9 — Wave–particle duality & electron diffraction [8 marks]

(a)(i) [2] Wave–particle duality: an electromagnetic wave can behave like a particle, and a moving particle can behave like a wave.

(a)(ii) [2] $\lambda = h/p$, where h is the Planck constant and p is the momentum of the particle.

(b)(i) [2] The pattern is similar to the diffraction of light, and diffraction is characteristic of wave behaviour — the rings show constructive and destructive interference. This is evidence of a wave nature for the electrons.

(b)(ii) [2] The de Broglie wavelength decreases ($\lambda = h/p$ and p increases), so the rings become closer together.

9702/42/O/N/25 Q8 — Hydrogen energy levels — line spectra & photon energies [10 marks]

(a) [3] Any three of: an electron moving between levels emits a single photon; the energy of the photon equals the difference between the energy levels; the photon energy depends on frequency ($E = hf$); the spectrum has discrete frequencies, so the differences between electron energies (and hence the levels themselves) must be discrete.

(b)(i) [1] $E = -13.6 \times 1.60 \times 10^{-19} = -2.18 \times 10^{-18}$ J

(b)(ii) [2] $\Delta E = hf = (6.63 \times 10^{-34} \times 2.47 \times 10^{15})/(1.60 \times 10^{-19})$
$= 10.2$ eV

(b)(iii) [4] n = 2 level = −13.6 + 10.2 = −3.4 eV.
n = 3 energy difference = 12.1 eV, so n = 3 = −13.6 + 12.1 = −1.5 eV.
n = 4 energy difference = 12.8 eV, so n = 4 = −13.6 + 12.8 = −0.8 eV.

9702/44/O/N/25 Q9 — Photoelectric effect — threshold frequency & work function [8 marks]

(a) [2] The emission of electrons from a metal surface when electromagnetic radiation is incident on it.

(b)(i) [1] threshold frequency of Y = 2F (work function $2\Phi = h \times 2F$).

(b)(ii) [1] Planck constant = Φ/F (since $\Phi = hF$).

(c) [4] Two diagonal straight lines with equal positive gradients (gradient = h) in the positive EK region only. Line X passes through (F, 0) and line Y through (2F, 0); X extrapolates back to (0, −Φ) and Y to (0, −2Φ).

9702/42/M/J/26 Q9 — Electron accelerated by 3.5 kV — speed & de Broglie wavelength [10 marks]

(a) [1] The de Broglie wavelength is the wavelength associated with a moving particle ($\lambda = h/p$).

(b) [4] Wave nature of matter: electron diffraction (or electron interference);
particle nature of matter: states of matter / state changes / gas behaviour / quantisation of charge / radioactive decay / Rutherford scattering / Brownian motion;
wave nature of EM radiation: diffraction / interference / polarisation;
particle nature of EM radiation: photoelectric effect (or emission/absorption line spectra, Compton scattering).

(c)(i) [3] $qV = \frac{1}{2}mv^2$
$1.60\times10^{-19}\times3500 = \frac{1}{2}\times9.11\times10^{-31}\times v^2$
$v^2 = 2\times1.60\times10^{-19}\times3500/(9.11\times10^{-31}) = 1.23\times10^{15}$
$v = 3.5\times10^7$ m s^{-1}

(c)(ii) [2] $\lambda = h/(mv) = 6.63\times10^{-34}/(9.11\times10^{-31}\times3.5\times10^7)$
$= 2.1\times10^{-11}$ m

9702/43/M/J/26 Q8 — X-rays for medical diagnosis — photon energy & wavelength [13 marks]

(a) [2] A photon is a quantum (packet) of energy of electromagnetic radiation — a discrete amount of electromagnetic energy, of energy $E = hf$.

(b)(i) [1] Electrons.

(b)(ii) [2] The electrons are accelerated to high speed and then decelerated rapidly by the metal target. Some of their kinetic energy is converted into the energy of photons, which are emitted as X-rays.

(b)(iii) [3] In a single interaction most electrons lose only part of their kinetic energy — some energy is dissipated as thermal energy in the target — and the amount lost varies from electron to

electron, so photons are produced with a range of energies (and hence wavelengths, since $\lambda = hc/E$ and wavelength is inversely proportional to photon energy).

The minimum wavelength occurs when all of an electron's kinetic energy becomes the energy of a single photon ($E = eV$).

(b)(iv) [3] $eV = hc/\lambda \rightarrow \lambda = hc/eV$

$\lambda = (6.63\times10^{-34} \times 3.00\times10^{8})/(1.60\times10^{-19} \times 52.0\times10^{3}) = 1.99\times10^{-25}/8.32\times10^{-15}$

$\lambda = 2.39\times10^{-11}$ m

(c) [2] Contrast depends on the linear attenuation (absorption) coefficients of the two materials: the greater the difference in attenuation coefficient, the greater the difference between the intensities transmitted through them.

Good contrast requires the transmitted intensities to be very different — if the materials have similar attenuation coefficients, the transmitted intensities are similar and the contrast is poor.

9702/44/M/J/26 Q8 — Photoelectric effect — Planck constant & work function energy [11 marks]

(a)(i) [1] The photoelectric effect (photoelectric emission).

(a)(ii) [1] The threshold frequency is the minimum frequency of radiation needed to cause emission of electrons from the metal surface.

(a)(iii) [3] Any three of: electrons need a minimum energy (the work function energy) to leave the surface; photon absorbed by an electron must have enough energy for the electron to escape; photons ar packets (quanta) of energy proportional to frequency ($E = hf$); a minimum photon energy therefo requires a minimum frequency, so radiation below the threshold frequency cannot eject electrons however intense it is.

(b)(i) [3] $hf = \Phi + E_K$ for each wavelength; subtracting the two equations:

$hc(1/\lambda_1 - 1/\lambda_2) = E_2 - E_1$

$h\times3.00\times10^{8}\times(1/178\times10^{-9} - 1/254\times10^{-9}) = (5.27-1.91)\times10^{-19}$

$h\times3.00\times10^{8}\times1.68\times10^{6} = 3.36\times10^{-19}$

$h = 6.66\times10^{-34}$ J s

(b)(ii) [3] $\Phi = hc/\lambda - E_K = 6.66\times10^{-34}\times3.00\times10^{8}/(178\times10^{-9}) - 5.27\times10^{-19}$

$= 1.12\times10^{-18} - 5.27\times10^{-19} = 5.95\times10^{-19}$ J

$\Phi = 5.95\times10^{-19}/1.60\times10^{-19} = 3.7$ eV

23 Nuclear physics

9702/42/F/M/24 Q8 — Nuclear fission of uranium-235 — binding energy & half-life [9 marks]

(a) [2] The binding energy of a nucleus is the minimum energy required to separate the nucleons (protons and neutrons) of the nucleus to infinity — equivalently, the energy released when the nucleons come together to form the nucleus.

(b)(i) [1] 235 + 1 = 142 + 90 + n → n = 4 neutrons

(b)(ii) [2] Energy released = (142×8.37 + 90×8.72) − (235×7.59)
= 1973 − 1784 = 190 MeV

(b)(iii) [1] Xenon-142 has too many neutrons (it is neutron-rich), so the nucleus is unstable (alternatively: its binding energy per nucleon is too small).

(b)(iv) [3] Decayed:undecayed = 31:1, so the fraction undecayed $N/N_0 = 1/32 = (½)^5$ — 5 half-lives have elapsed in 6.0 s:
$t_{½} = 6.0/5 = 1.2$ s

9702/41/M/J/24 Q9 — Radioactive decay of isotope X — half-life & nucleon number [8 marks]

(a) [1] The time taken for the activity of the sample to halve.

(b)(i) [1] The activity of X at time t — the magnitude of the gradient of the number–time graph, since the number of X nuclei is decreasing (activity = −dN/dt).

(b)(ii) [3] Any three of:
1. Y is a stable isotope (its number of nuclei eventually becomes constant);
2. the total number of nuclei is constant ($X + Y = 4.0\times10^{22}$);
3. the half-life of X is 13.6 s;
4. the decay constant of X is $\lambda = 0.693/13.6 = 0.051\ s^{-1}$;
5. the amount of X at t = 0 is $4.0\times10^{22}/6.02\times10^{23} = 0.066$ mol;
6. the activity of X at t = 0 is $\lambda N = 0.051 \times 4.0\times10^{22} = 2.0\times10^{21}$ Bq.

(c) [3] Number of nuclei at t = 0: $N_0 = 4.0\times10^{22}$ (from Fig. 9.1).
Mass of one nucleus = $7.3\times10^{-4} / 4.0\times10^{22} = 1.8\times10^{-26}$ kg
Nucleon number = mass of nucleus / $(1.66\times10^{-27}) = 1.8\times10^{-26} / 1.66\times10^{-27} = 11$

9702/42/M/J/24 Q9 — Binding energy of polonium-212 — mass defect & alpha-decay energy [13 m

(a) [2] The binding energy of a nucleus is the energy required to separate all the nucleons in the nucleus to infinity (so that they are no longer interacting).

(b)(i) [3] Po-212 has Z = 84, so 84 protons and 212 − 84 = 128 neutrons.
$\Delta m = (84 \times 1.007276 + 128 \times 1.008665) - 211.942749$
$= 213.7203 - 211.9427 = 1.778$ u
$= 1.778 \times 1.66\times10^{-27}$
$= 2.95\times10^{-27}$ kg

(b)(ii) [2] $E = \Delta mc^2 = 2.95\times10^{-27} \times (3.00\times10^{8})^2$
$= 2.66\times10^{-10}$ J

(b)(iii) [1] Binding energy per nucleon = $2.66\times10^{-10} / 212$
$= 1.25\times10^{-12}$ J

(c)(i) [2] Sketch a curve that rises steeply from A = 1 to a single peak at about A = 50–60, then decreases continuously with a shallow negative gradient towards A = 250.

(c)(ii) [1] Place an X on the curve on the right-hand descending branch, at A ≈ 212 (to the right of the peak).

(c)(iii) [2] The daughter nucleus has a lower nucleon number than polonium-212, so, reading from Fig. 9.1, has a greater binding energy per nucleon (it lies further up the curve).

Energy is released because the products of the decay are more tightly bound, so the alpha-decay releases energy.

9702/42/O/N/24 Q10 — Random decay, fission & fusion — binding energy per nucleon [8 marks]

(a)(i) [1] Random: it cannot be predicted when a particular nucleus will decay (or which nucleus will decay next).

(a)(ii) [1] Spontaneous: the decay is not affected by external (environmental) factors — it occurs on its own.

(a)(iii) [1] Evidence: fluctuations in the (measured) count rate.

(b)(i) [3] Any three of: large nuclei undergo fission whereas small nuclei undergo fusion; fission involves one nucleus splitting into two (or more) smaller nuclei; fusion involves two nuclei joining together to form one (larger) nucleus; fission is usually initiated by neutron bombardment; fusion is usually initiated by (very) high temperatures.

(b)(ii) [2] Binding energy per nucleon is greatest for intermediate nucleon numbers. Fission (large → intermediate nuclei) and fusion (small → intermediate nuclei) both move towards greater binding energy per nucleon, so energy is released in both processes.

9702/42/F/M/25 Q9 — Polonium-193 decay — binding energy, half-life & PET [10 marks]

(a)(i) [1] Random — one cannot predict when a particular nucleus will decay (or which nucleus will decay next).

(a)(ii) [1] Spontaneous — the decay is not affected by external or environmental factors.

(b) [1] The time for the activity (or number of nuclei) to halve.

(c) [2] Energy released = total binding energy of products − total binding energy of the parent nucleus:
= (189 × 7.826) + (4 × 7.074) − (193 × 7.774)
= 1479.1 + 28.3 − 1500.4 = 7.03 eV

(d)(i) [1] The decay constant (the magnitude of the gradient of the $\ln(N/N_0)$–t graph equals λ).

(d)(ii) [2] Decay constant λ = magnitude of gradient = 1.4/0.84.
Half-life = ln 2/λ = ln 2/(1.4/0.84)
= 0.42 ms

(e)(i) [1] The emitted positrons collide with electrons and annihilate, producing gamma-ray photons.

(e)(ii) [1] The half-life is long enough to allow the investigation to be carried out, but not so long that the patient is exposed to radiation unnecessarily.

9702/41/M/J/25 Q9 — Radioactive decay — activity, decay constant & half-life [10 marks]

(a) [1] The number of nuclear disintegrations per unit time.

(b) [3] The activity is proportional to the number of undecayed nuclei (A = λN), and the activity equals the rate of decrease of N, so dN/dt = −λN — the number of nuclei falls at a rate proportional to itself, giving an exponential variation.

(c)(i) [2] $A = A_0e^{(-\lambda t)}$, so $120 = 180e^{(-\lambda \times 8.4)}$
$e^{(-8.4\lambda)} = 120/180 = 0.667$
$\lambda = -\ln 0.667/8.4 = 0.048\ \text{min}^{-1}$

(c)(ii) [1] Half-life = ln 2/λ = ln 2/0.048 = 14 min

(c)(iii) [3] An exponential decay curve starting at (0, 180 Bq), with a negative gradient that decreases in magnitude, passing through (8.4, 120) and (14, 90), and approaching the t-axis by t = 24 min.

9702/42/M/J/25 Q10 — Radioactive decay — half-life, decay constant & activity [9 marks]

(a) [1] Decay that is not affected by external/environmental factors — it occurs of its own accord.

(b)(i) [3] From Fig. 10.1, X halves from 4A to 2A at t = 2T and Y halves from A to ½A at t = 3T, so half-life of X = 2T and half-life of Y = 3T.
Decay constant λ = ln 2/half-life: λX = ln 2/2T, λY = ln 2/3T.
Initial number $N_0 = A_0/\lambda$: $N_0(X) = 4A/(\ln 2/2T) = 8AT/\ln 2$ and $N_0(Y) = A/(\ln 2/3T) = 3AT/\ln 2$.

(b)(ii) [3] Equal activities: $4A\ e^{(-\lambda Xt)} = A\ e^{(-\lambda Yt)}$
$4 = e^{(t \ln 2(1/2T - 1/3T))} = e^{(t \ln 2/6T)}$
ln 4 = t ln 2/6T → 2 ln 2 = t ln 2/6T
t = 12T

(c) [2] The count rate is less than the activity because the radiation is emitted in all directions (not just towards the detector), some is absorbed by the air/sample/window of the detector, and some radiation may not register even though it reaches the detector.

9702/43/M/J/25 Q9 — Radioactive decay — activity, decay constant & half-life [10 marks]

(a) [1] The number of nuclear disintegrations per unit time.

(b) [3] The activity is proportional to the number of undecayed nuclei (A = λN), and the activity equals the rate of decrease of N, so dN/dt = −λN — the number of nuclei falls at a rate proportional to itself, giving an exponential variation.

(c)(i) [2] $A = A_0e^{(-\lambda t)}$, so $120 = 180e^{(-\lambda \times 8.4)}$
$e^{(-8.4\lambda)} = 120/180 = 0.667$
$\lambda = -\ln 0.667/8.4 = 0.048\ \text{min}^{-1}$

(c)(ii) [1] Half-life = ln 2/λ = ln 2/0.048 = 14 min

(c)(iii) [3] An exponential decay curve starting at (0, 180 Bq), with a negative gradient that decreases in magnitude, passing through (8.4, 120) and (14, 90), and approaching the t-axis by t = 24 min.

9702/44/M/J/25 Q11 — Nuclear fusion in a star — energy, luminosity & temperature [9 marks]

(a)(i) [1] Nuclear fusion.

(a)(ii) [3] Δm = 0.030377 − (3 × 0.002388) = 0.023213 u (the mass difference between the helium-4 and t three deuterium nuclei).

$E = \Delta m\, c^2 = 0.023213 \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$

$= 3.47 \times 10^{-12}$ J

(b)(i) [3] Mass of 1 mol of helium-4 = 4×10^{-3} kg. Number of helium-4 nuclei produced per second $= (6.02 \times 10^{23} \times 7.34 \times 10^{11})/(4 \times 10^{-3}) = 1.10 \times 10^{38}$ s^{-1}.

Luminosity = (number per second) × (energy per nucleus) $= 1.10 \times 10^{38} \times 3.47 \times 10^{-12}$

$= 3.83 \times 10^{26}$ W

(b)(ii) [2] $L = 4\pi\sigma r^2 T^4$

$3.83 \times 10^{26} = 4\pi \times 5.67 \times 10^{-8} \times (6.96 \times 10^8)^2 \times T^4$

T = 5770 K

9702/41/O/N/25 Q8 — Uranium-238 alpha decay — mass, gamma wavelength & neutrinos [11 ma

(a) [2] A packet/quantum of energy of electromagnetic radiation.

(b)(i) [3] $E = \Delta m\, c^2$, so $\Delta m = E/c^2 = (4.274 \times 10^6 \times 1.60 \times 10^{-19})/(1.66 \times 10^{-27} \times (3.00 \times 10^8)^2) = 0.00458$ u.

m = 233.915174 + 4.000407 + 0.00458

= 237.92016 u

(b)(ii) [3] The gamma photon carries the remaining energy: 4.274 − 4.200 = 0.074 MeV.

$E = hc/\lambda$, so $\lambda = hc/E = (6.63 \times 10^{-34} \times 3.00 \times 10^8)/(0.074 \times 1.60 \times 10^{-13})$

$= 1.7 \times 10^{-11}$ m

(b)(iii) [1] The true energy of the gamma photon is smaller (the thorium nucleus carries some kinetic energy), so the true wavelength is larger than the calculated value.

(c) [2] Beta decay also emits (anti)neutrinos, and the particles emitted in beta decay carry varying amounts of energy, so the energy of the gamma photon is variable between decays — there is no single wavelength.

9702/42/O/N/25 Q9 — Nuclear fusion & Sirius — mass defect, luminosity & temperature [13 mark

(a) [2] The difference between the mass of the nucleus and the mass of its constituent nucleons (when the nucleons are separated to infinity).

(b) [4] Δm = (2 × 2.013553) − 4.001505 = 0.025601 u.

Energy per helium-4 nucleus $= \Delta m\, c^2 = 0.025601 \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2 = 3.82 \times 10^{-12}$ J.

Energy for 1.00 mol $= 3.82 \times 10^{-12} \times 6.02 \times 10^{23} = 2.30 \times 10^{12}$ J

(c)(i) [2] $L = mc^2$ (mass lost per second) $= 1.09 \times 10^{11} \times (3.00 \times 10^8)^2$

$= 9.81 \times 10^{27}$ W

(c)(ii) [2] $L = 4\pi\sigma r^2 T^4$

$9.81 \times 10^{27} = 4\pi \times 5.67 \times 10^{-8} \times (1.19 \times 10^9)^2 \times T^4$

T = 9930 K

(d) [3] Standard candles have a known luminosity. Measure the radiant flux intensity from the galaxy on the Earth, then use $F = L/(4\pi d^2)$ to calculate the distance d.

9702/43/O/N/25 Q8 — Uranium-238 alpha decay — mass, gamma wavelength & neutrinos [11 ma

(a) [2] A packet/quantum of energy of electromagnetic radiation.

(b)(i) [3] $E = \Delta m\, c^2$, so $\Delta m = E/c^2 = (4.274 \times 10^6 \times 1.60 \times 10^{-19})/(1.66 \times 10^{-27} \times (3.00 \times 10^8)^2) = 0.00458$ u.

m = 233.915174 + 4.000407 + 0.00458

= 237.92016 u

(b)(ii) [3] The gamma photon carries the remaining energy: 4.274 − 4.200 = 0.074 MeV.

$E = hc/\lambda$, so $\lambda = hc/E = (6.63 \times 10^{-34} \times 3.00 \times 10^8)/(0.074 \times 1.60 \times 10^{-13})$

$= 1.7 \times 10^{-11}$ m

(b)(iii) [1] The true energy of the gamma photon is smaller (the thorium nucleus carries some kinetic energy), so the true wavelength is larger than the calculated value.

(c) [2] Beta decay also emits (anti)neutrinos, and the particles emitted in beta decay carry varying amounts of energy, so the energy of the gamma photon is variable between decays — there is no single wavelength.

9702/44/O/N/25 Q8 — Oxygen-15 — radioactive decay & PET scanning [15 marks]

(a) [2] A tracer is a radioactive substance introduced into the body that is absorbed by the tissues being studied.

(b)(i) [2] P = 15, Q = 7, R = 0, S = +1 ($^{15}_{8}O \rightarrow {}^{15}_{7}N + \beta^{+} + \nu$).

(b)(ii) [1] An (electron) neutrino.

(c)(i) [1] The number of nuclear disintegrations per unit time.

(c)(ii) [2] $\lambda = \ln 2/t_{½} = \ln 2/(2.04 \times 60)$
$= 5.66 \times 10^{-3}\ s^{-1}$

(c)(iii) [4] N = mass/mass of one atom = $2.85 \times 10^{-6}/(15 \times 1.66 \times 10^{-27})$.
$A = \lambda N = 5.66 \times 10^{-3} \times 2.85 \times 10^{-6}/(15 \times 1.66 \times 10^{-27})$
$= 6.48 \times 10^{17}\ s^{-1}$

(d)(i) [1] Gamma-ray photons.

(d)(ii) [2] The positron collides with an electron in the body, and annihilation converts their masses into the energy of the gamma photons.

9702/41/M/J/26 Q9 — Thorium-228 decay — mass defect & alpha emission [9 marks]

(a) [2] The mass defect of a nucleus is the difference between the mass of the nucleus and the sum of the masses of its constituent nucleons (protons and neutrons), when the nucleons are separated to infinity.

(b)(i) [4] α-decay produces Ra-224 (plus He-4); β^{+} decay would produce Ac-228 and β^{-} decay would produc Pa-228.

Total mass defect of the α-decay products = 1.846 840 + 0.030 377 = 1.877 217 u, which is greater than the mass defect of thorium-228 (1.871 290 u), so α-decay releases energy.
The mass defects of Ac-228 (1.869 820 u) and Pa-228 (1.866 129 u) are both smaller than that of thorium-228, so β^{+} and β^{-} decay would not release energy. A decay can occur only if the mass defect of the products is greater than that of the parent nucleus — hence thorium-228 is an α-emitter and not a β^{-} or β^{+} emitter.

(b)(ii) [3] $\Delta m = 1.877\ 217 - 1.871\ 290 = 0.005\ 927$ u.
$E = \Delta mc^2 = 0.005\ 927 \times 1.66 \times 10^{-27} \times (3.00 \times 10^{8})^2 = 8.85 \times 10^{-13}$ J
$= 8.85 \times 10^{-13}/(1.60 \times 10^{-13}) = 5.53$ MeV

9702/42/M/J/26 Q10 — Star fusing hydrogen to helium — binding energy & mass loss [11 marks]

(a) [2] The binding energy of a nucleus is the energy required to separate all the nucleons (protons and neutrons) in the nucleus to infinity — equivalently, the energy released when the nucleons come together to form the nucleus.

(b)(i) [1] $4\ {}^{1}H \rightarrow {}^{4}He + 2X$, so X is the positron ($\beta^{+}$).

(b)(ii) [2] $E = \Delta mc^2 = 0.030377 \times 1.66 \times 10^{-27} \times (3.00 \times 10^{8})^2$
$= 4.54 \times 10^{-12}$ J

(c)(i) [2] $F = L/(4\pi d^2) = 7.98 \times 10^{27}/(4\pi \times (8.15 \times 10^{16})^2)$
$= 9.56 \times 10^{-8}\ W\ m^{-2}$

(c)(ii) [4] Number of He-4 nuclei formed per second = L/E_He = $7.98 \times 10^{27}/4.54 \times 10^{-12} = 1.76 \times 10^{39}\ s^{-1}$. Eac He-4 nucleus is formed from 4 hydrogen nuclei, each of mass 1.67×10^{-27} kg:
mass rate = $4 \times 1.67 \times 10^{-27} \times 7.98 \times 10^{27}/4.54 \times 10^{-12}$
$= 1.17 \times 10^{13}\ kg\ s^{-1}$

9702/43/M/J/26 Q9 — Thorium-228 — mass defect & energy released by decay [9 marks]

(a) [2] The mass defect of a nucleus is the difference between the mass of the nucleus and the sum of the masses of its constituent nucleons (protons and neutrons) when separated to infinity.

(b)(i) [4] α-decay: $^{228}_{90}Th \rightarrow {}^{224}_{88}Ra + {}^{4}_{2}He$; the total mass defect of the products = 1.846840 + 0.030377 = 1.877217 u.

β^{-}-decay would give $^{228}_{91}Pa$ (mass defect 1.866129 u); β^{+}-decay would give $^{228}_{89}Ac$ (mass defect 1.869820 u).
Decay is possible only if the mass defect of the products is greater than that of the parent nucleus (1.871290 u), i.e. if energy is released.

Only the α-decay products satisfy this condition, so thorium-228 is an α-emitter and not a β^-- or β^+-emitter.

(b)(ii) [3] $E = \Delta mc^2$

$\Delta m = 1.877217 - 1.871290 = 0.005927$ u $= 0.005927 \times 1.66\times10^{-27}$ kg

$E = 0.005927 \times 1.66\times10^{-27} \times (3.00\times10^{8})^2 = 8.85\times10^{-13}$ J

$E = 8.85\times10^{-13}/(1.60\times10^{-13}) = 5.53$ MeV (to 3 s.f.)

24 Medical physics

9702/42/F/M/24 Q9 — X-ray imaging & ultrasound — attenuation and acoustic impedance [13 ma

(a)(i) [2] eV = hc/λ → λ = hc/(eV) = $6.63\times10^{-34}\times3.00\times10^{8}/(1.6\times10^{-19}\times84\times10^{3})$
$= 1.5\times10^{-11}$ m

(a)(ii) [2] A large proportion of the kinetic energy of the electrons is converted to thermal energy in the target (and some X-rays are absorbed by it), so the target becomes very hot. Tungsten is used because its high melting point (3420 °C) prevents the target from melting.

(b) [3] $I = I_0e^{(-\mu x)}$: total attenuation coefficient $\mu = 0.22 + 3.0 = 3.22$ cm^{-1}
$0.13 = e^{(-3.22x)}$
$x = -\ln(0.13)/3.22 = 0.63$ cm

(c)(i) [2] The specific acoustic impedance Z of a medium is the product of the density of the medium and the speed of ultrasound in the medium: $Z = \rho c$.

(c)(ii) [2] $I_R/I_0 = (Z_2-Z_1)^2/(Z_2+Z_1)^2 = (7.8-1.7)^2/(7.8+1.7)^2 = 6.1^2/9.5^2 = 0.41$ (reflected)
fraction transmitted = 1 − 0.41 = 59%

(c)(iii) [2] Any two of: the beam crosses more than one boundary (tissue–bone and bone–tissue), so there a more reflections, each reducing the transmitted intensity further; the ultrasound is attenuated (absorbed and scattered) as it travels through the matter.

9702/42/M/J/24 Q10 — Ultrasound imaging — acoustic impedance & intensity reflection coefficie marks]

(a) [2] The time between emission and detection of a reflected pulse gives the depth (position) of the boundary between the two tissues.
The intensity (amplitude) of the reflected pulse gives information about the nature of the boundary (the acoustic-impedance mismatch, and hence the different tissues involved).

(b)(i) [2] The specific acoustic impedance of a medium is the product of the density of the medium and the speed of (ultra)sound in the medium: $Z = \rho c$.

(b)(ii) [3] $Z_{water} = 1000 \times 1420 = 1.42\times10^{6}$ kg m^{-2} s^{-1}
$Z_{glass} = 2500 \times 4560 = 11.4\times10^{6}$ kg m^{-2} s^{-1}
$I_R/I_0 = (Z2 - Z1)^2 / (Z2 + Z1)^2 = (11.4 - 1.42)^2 / (11.4 + 1.42)^2$
$= 9.98^2 / 12.82^2 = 99.6 / 164.4$
$= 0.61$

9702/41/O/N/24 Q9 — Fluorine-18 tracer — decay, activity & PET imaging [10 marks]

(a)(i) [1] The positron (the antiparticle of the electron).

(a)(ii) [1] $\lambda = \ln 2/t_{½} = 0.693/(110 \times 60) = 0.693/6600 = 1.05\times10^{-4}$ s^{-1}

(a)(iii) [3] $N = M/(18u) = 2.1\times10^{-12}/(18 \times 1.66\times10^{-27}) = 7.0\times10^{13}$ nuclei
$A = \lambda N = 1.05\times10^{-4} \times 7.0\times10^{13} = 7.4\times10^{9}$ Bq

(b)(i) [3] Any three of:
1. the β^+ particle (positron) annihilates when it interacts with an electron in the body;
2. the mass of the two particles is converted into energy;
3. two gamma-ray photons are produced, travelling in opposite directions, and leave the body;
4. the difference in the arrival times of the two photons at the detector is processed to locate the source and build the image.

(b)(ii) [2] With a shorter half-life, the sample would decay (almost) completely before the scan is complete, so there would be insufficient activity to form the image. With a longer half-life, the patient would be exposed to harmful (ionising) radiation for longer than necessary (or a larger dose would be needed to give a detectable activity) — so 110 minutes is long enough for the scan but short enough to limit exposure.

9702/44/M/J/25 Q10 — Ultrasound — piezoelectric transducer, impedance & attenuation [9 mark

(a) [3] An alternating p.d. is applied across the crystal, causing it to vibrate. When the frequency of the applied p.d. equals the natural frequency of the crystal it resonates, and this natural frequency is in the ultrasound range — so ultrasound waves are generated.

(b)(i) [2] Specific acoustic impedance is the product of the density of the medium and the speed of (ultra)sound in the medium ($Z = \rho c$).

(b)(ii) [2] The reflected fraction α is close to 0 when Z_1 and Z_2 are equal, and close to 1 when Z_1 and Z_2

are very different.

(c) [2] I = $I_0 e^{(-\mu x)}$, so $\ln(I/I_0) = -\mu x$.

$\ln 0.62 = -\mu \times 2.1$

$\mu = 0.23\ cm^{-1}$

9702/41/O/N/25 Q10 — Ultrasound — acoustic impedance & reflection coefficient [8 marks]

(a) [2] The product of the density of the medium and the speed of sound in the medium ($Z = \rho c$).

(b) [2] The ultrasound waves cause the crystal to vibrate, and these vibrations generate an induced e.m.f. across the crystal, which is then detected.

(c)(i) [2] Intensity reflection coefficient = $(Z_1 - Z_2)^2/(Z_1 + Z_2)^2$

$= (40.4 - 1.48)^2/(40.4 + 1.48)^2 = 0.86$

(c)(ii) [2] The Z values of body tissue and water are very similar, so almost all the ultrasound is transmitted (almost none is reflected).

9702/42/O/N/25 Q10 — X-rays — attenuation, contrast & CT scanning [9 marks]

(a) [1] Contrast is the difference in the degrees of blackening between different parts of the image.

(b)(i) [1] Region B passes through 5.8 cm of P only:

$I = I_0 e^{(-\mu x)} = I_0 e^{(-0.35 \times 5.8)} = 0.13\ I_0$

(b)(ii) [3] Region A passes through (5.8 − 2.1) = 3.7 cm of P and 2.1 cm of Q:

$0.053 I_0 = I_0 e^{(-(0.35 \times 3.7 + 2.1\mu))}$

$0.35 \times 3.7 + 2.1\mu = -\ln 0.053 = 2.94$

$\mu = 0.78\ cm^{-1}$

(b)(iii) [1] The two emerging intensities differ by only a factor of about 2.5 (0.13 vs 0.053), so the contrast between the regions is poor.

(c) [3] The structure is scanned in thin sections, with many scans taken from different angles around each section. The scanning is repeated for all sections and the data are compiled to form a 3D image.

9702/43/O/N/25 Q10 — Ultrasound — acoustic impedance & reflection coefficient [8 marks]

(a) [2] The product of the density of the medium and the speed of sound in the medium ($Z = \rho c$).

(b) [2] The ultrasound waves cause the crystal to vibrate, and these vibrations generate an induced e.m.f. across the crystal, which is then detected.

(c)(i) [2] Intensity reflection coefficient = $(Z_1 - Z_2)^2/(Z_1 + Z_2)^2$

$= (40.4 - 1.48)^2/(40.4 + 1.48)^2 = 0.86$

(c)(ii) [2] The Z values of body tissue and water are very similar, so almost all the ultrasound is transmitted (almost none is reflected).

9702/41/M/J/26 Q8 — X-ray production — photons, minimum wavelength & contrast [13 marks]

(a) [2] A photon is a packet (quantum) of energy of electromagnetic radiation, carrying energy $E = hf$.

(b)(i) [1] Electrons.

(b)(ii) [2] The electrons are decelerated by the metal target. (Some of) the kinetic energy of the electrons becomes the energy of photons, which are emitted as X-rays.

(b)(iii) [3] Some of an electron's kinetic energy is dissipated as thermal energy in the target, so the photon energy varies from electron to electron — giving a range of wavelengths. The wavelength is inversely proportional to the photon energy ($\lambda = hc/E$). The minimum wavelength occurs when all of an electron's kinetic energy becomes the energy of a single photon.

(b)(iv) [3] $eV = hc/\lambda_{min}$, so $\lambda_{min} = hc/(eV)$

$= (6.63 \times 10^{-34} \times 3.00 \times 10^8)/(1.60 \times 10^{-19} \times 52.0 \times 10^3) = 1.99 \times 10^{-25}/8.32 \times 10^{-15}$

$= 2.39 \times 10^{-11}$ m

(c) [2] The image contrast is determined by the (linear) attenuation coefficients μ of the two materials — how strongly each material attenuates the X-ray beam ($I = I_0 e^{(-\mu x)}$). Good contrast requires the two materials to have very different attenuation coefficients, so that the transmitted intensities through them are very different.

9702/44/M/J/26 Q9 — Ultrasound transducer — acoustic impedance & attenuation of materials [8 marks]

(a) [3] A p.d. applied across the piezo-electric crystal causes it to distort (change shape); an

alternating p.d. is applied, so the crystal vibrates; the vibrations of the crystal at the ultrasound frequency cause oscillations in the surrounding medium, generating the ultrasound waves.

(b)(i) [2] $Z = \rho c = 8960 \times 4720 = 4.23 \times 10^{7}$ kg m^{-2} s^{-1}

(b)(ii) [3] Any three of: the total thickness of the structure is 12.0 cm, with material X occupying 10.0 cm along PQ and material Y 2.0 cm thick (X is thicker than Y); the linear attenuation coefficient of X is 3.2 cm^{-1} and that of Y is 1.5 cm^{-1}, so X attenuates ultrasound more strongly than Y; the intensity transmission coefficient at the X–Y boundary is 0.37, so the intensity reflection coefficient there is 0.63.

25 Astronomy and cosmology

9702/42/F/M/24 Q10 — The Sun & a receding galaxy — radius, flux & redshift [10 marks]

(a)(i) [2] $L = 4\pi\sigma r^2T^4 \rightarrow r^2 = L/(4\pi\sigma T^4)$
$r^2 = 3.85\times10^{26}/(4\pi\times5.67\times10^{-8}\times5780^4) = 4.84\times10^{17}$
$r = 6.96\times10^8$ m

(a)(ii) [2] $F = L/(4\pi d^2) = 3.85\times10^{26}/(4\pi\times(1.50\times10^{11})^2) = 1.36\times10^3$ W m^{-2}

(a)(iii) [2] A curve of the same black-body shape that is always below the Sun's curve (same radius but lowe temperature gives lower luminosity) with its peak intensity at a greater wavelength (Wien's law: $\lambda_{max} \propto 1/T$).

(b)(i) [1] The same pattern of five lines (at 397, 410, 434, 486 and 656 nm), all shifted to longer wavelengths by similar amounts (redshift).

(b)(ii) [2] Lowest visible energy emission = the longest visible wavelength, 656 nm:
$\Delta\lambda/\lambda = v/c \rightarrow \Delta\lambda = 656\times(21\,400/300\,000) = 46.8$ nm
observed $\lambda = 656 + 46.8 = 703$ nm

(b)(iii) [1] The value is too low — the observed wavelength is increased by the redshift, and Wien's law ($\lambda_{max} \propto 1/T$) then gives a temperature that is too low.

9702/41/M/J/24 Q10 — Luminosity of the Sun — Stefan–Boltzmann & Wien's laws [9 marks]

(a)(i) [2] The luminosity of a star is the total power of radiation emitted by the star (over all wavelengths).

(a)(ii) [3] A standard candle is an object of known luminosity L. The radiant flux intensity F received from it is measured by the observer, and since $F = L/(4\pi d^2)$, the distance of the galaxy can be determined: $d = \sqrt{(L/4\pi F)}$.

(b)(i) [2] $L = 4\pi\sigma r^2T^4 = 4\pi \times 5.67\times10^{-8} \times (6.96\times10^8)^2 \times 5780^4$
$= 3.85\times10^{26}$ W

(b)(ii) [2] $\lambda_{max}T = \text{constant} \rightarrow T = (5780 \times 501) / 624 = 4640$ K

9702/42/M/J/24 Q8 — Hydrogen spectrum from a receding galaxy — redshift & Hubble distance [marks]

(a)(i) [2] The motion of the galaxy away from the Earth causes a redshift: the observed frequency of the lines is lower (and the observed wavelength longer) than the emitted values, so all the lines are shifted towards the red (low-frequency) end of the spectrum.

(a)(ii) [2] Draw the three lines to the left of the corresponding printed lines, with each drawn line displaced from its printed line by approximately the same distance (same $\Delta\lambda/\lambda$ for all three lines).

(b)(i) [2] $E = hf = hc/\lambda = (6.63\times10^{-34} \times 3.00\times10^8) / (488\times10^{-9})$
$= 4.08\times10^{-19}$ J

(b)(ii) [2] Photon energy $= 4.08\times10^{-19} / 1.60\times10^{-19} = 2.55$ eV
$hf = E1 - E2$, so $E1 = -3.40 + 2.55$
$= -0.85$ eV

(b)(iii) [2] $\Delta\lambda/\lambda = v/c$
$\Delta\lambda = \lambda v/c = (488 \times 6.2\times10^6) / 3.00\times10^8 = 10.1 \approx 10$ nm
Observed wavelength $= 488 + 10$
$= 498$ nm

(c) [2] $v = H0d$
$d = v/H0 = 6.2\times10^6 / 2.3\times10^{-18}$
$= 2.7\times10^{24}$ m

9702/41/O/N/24 Q10 — Redshift of galaxy light — distance, speed & Hubble's law [10 marks]

(a) [3] Any three of:
1. redshift is the increase in the observed wavelength (decrease in observed frequency) of radiation from a source moving away from the observer (Doppler effect);
2. the radiation from distant galaxies is observed to be redshifted;
3. this provides evidence that the galaxies are moving away from the Earth (away from each other);
4. if the galaxies are moving apart, the Universe must be expanding.

(b)(i) [2] $F = L/4\pi d^2 \rightarrow d = \sqrt{(L/4\pi F)}$

$d = \sqrt{(1.90\times10^{36}/(4\pi \times 8.42\times10^{-16}))} = \sqrt{(1.80\times10^{50})} = 1.34\times10^{25}$ m

(b)(ii) [2] $\Delta\lambda/\lambda = v/c$

$v = (726 - 658)/658 \times 3.00\times10^{8} = 0.103 \times 3.00\times10^{8} = 3.1\times10^{7}$ m s^{-1}

(c)(i) [2] A straight line passing through the origin with a positive gradient — v is directly proportional to d (Hubble's law).

(c)(ii) [1] The Hubble constant (H_0).

9702/42/F/M/25 Q10 — The Sun — rotation, redshift & luminosity [11 marks]

(a)(i) [1] The total power of radiation emitted (by the star).

(a)(ii) [3] A standard candle has a known luminosity. Measure the radiant flux intensity F received at the Earth, then use $F = L/(4\pi d^2)$ to calculate the distance d to the galaxy.

(b)(i) [3] The two limbs of the Sun rotate in opposite directions, so the light from X and Y shows a rotational Doppler shift with $\Delta\lambda = 656.2877 - 656.2831$ nm. Using $\Delta\lambda/\lambda = v/c$ and $v = 2\pi R/T$:

$R = vT/2\pi = [3.00 \times 10^{8} \times (656.2877 - 656.2831)/656.2831] \times (2.07 \times 10^{6})/2\pi$

$= 6.93 \times 10^{8}$ m

(b)(ii) [2] Point Z is moving towards the Earth, so its light is blue-shifted: the observed wavelength from Z is less than the emitted wavelength.

(b)(iii) [2] $L = 4\pi\sigma r^2T^4$

$3.8 \times 10^{26} = 4\pi \times 5.67 \times 10^{-8} \times (6.93 \times 10^{8})^2 \times T^4$

$T = 5800$ K

9702/41/M/J/25 Q10 — Hubble's law & redshift — galaxy recession [10 marks]

(a) [2] The speed of recession of a galaxy is (directly) proportional to its distance from the observer ($v = H_0d$), where the speed is that of the galaxy receding from the observer and d is its distance from the observer.

(b)(i) [2] The observed wavelength is longer than the emitted wavelength (red-shifted), so the galaxy is receding from the Earth (the Doppler shift of the light).

(b)(ii) [2] $\Delta\lambda/\lambda = v/c$

$(4.91 - 4.62)/4.62 = v/(3.00 \times 10^{8})$

$v = 1.9 \times 10^{7}$ m s^{-1}

(b)(iii) [2] The wavelength of maximum intensity is inversely proportional to the temperature (Wien's law). The observed wavelength is too high (red-shifted), so the determined temperature is too low compared with the true value.

(c) [2] $v = H_0d$, so $d = v/H_0 = (1.9 \times 10^{7})/(2.3 \times 10^{-18})$

$= 8.3 \times 10^{24}$ m

9702/43/M/J/25 Q10 — Hubble's law & redshift — galaxy recession [10 marks]

(a) [2] The speed of recession of a galaxy is (directly) proportional to its distance from the observer ($v = H_0d$), where the speed is that of the galaxy receding from the observer and d is its distance from the observer.

(b)(i) [2] The observed wavelength is longer than the emitted wavelength (red-shifted), so the galaxy is receding from the Earth (the Doppler shift of the light).

(b)(ii) [2] $\Delta\lambda/\lambda = v/c$

$(4.91 - 4.62)/4.62 = v/(3.00 \times 10^{8})$

$v = 1.9 \times 10^{7}$ m s^{-1}

(b)(iii) [2] The wavelength of maximum intensity is inversely proportional to the temperature (Wien's law). The observed wavelength is too high (red-shifted), so the determined temperature is too low compared with the true value.

(c) [2] $v = H_0d$, so $d = v/H_0 = (1.9 \times 10^{7})/(2.3 \times 10^{-18})$

$= 8.3 \times 10^{24}$ m

9702/41/O/N/25 Q9 — Wien's law & luminosity — conclusions about star X [7 marks]

(a) [2] The temperature is inversely proportional to the wavelength at which the maximum emission occurs ($\lambda max \propto 1/T$), where T is the thermodynamic temperature of the star's surface.

(b) [3] Any three of: star X has a surface temperature of 7000 K (higher than the Sun's 5770 K); star X has a higher luminosity than the Sun; the luminosity of star X is 2.7×10^{27} W; the radius of star X is 1.3×10^{9} m.

(c) [2] The light from star X is red-shifted (the galaxy is receding from the Earth), so the wavelength of the peak emission would be greater than its true value.

9702/43/O/N/25 Q9 — Wien's law & luminosity — conclusions about star X [7 marks]

(a) [2] The temperature is inversely proportional to the wavelength at which the maximum emission occurs ($\lambda_{max} \propto 1/T$), where T is the thermodynamic temperature of the star's surface.

(b) [3] Any three of: star X has a surface temperature of 7000 K (higher than the Sun's 5770 K); star X has a higher luminosity than the Sun; the luminosity of star X is 2.7×10^{27} W; the radius of star X is 1.3×10^{9} m.

(c) [2] The light from star X is red-shifted (the galaxy is receding from the Earth), so the wavelength of the peak emission would be greater than its true value.

9702/44/O/N/25 Q10 — Redshift & Hubble's law — the expanding universe [7 marks]

(a) [2] Redshift is the increase in observed wavelength (decrease in observed frequency) of light caused by the recession of the galaxy from the observer.

(b) [2] Distant galaxies show redshift, so the galaxies are moving apart — which means the universe must be expanding.

(c) [3] Hubble's law states that the speed of recession is proportional to distance (more distant galaxies recede faster). More distant galaxies represent further back in time, so long ago all the matter in the universe must have been very close together (and moving apart very fast) — hence the Big Bang.

9702/41/M/J/26 Q10 — Sun's luminosity — radiant flux, Wien's law & star temperature [8 marks]

(a) [2] Wien's displacement law states that the (thermodynamic) temperature of the surface of a star is inversely proportional to the wavelength at which the maximum emission rate (intensity) occurs — that is, $\lambda_{max}T$ = constant.

(b)(i) [2] $L = 4\pi\sigma r^2T^4 = 4\pi \times 5.67 \times 10^{-8} \times (6.96 \times 10^8)^2 \times 5770^4$
$= 3.83 \times 10^{26}$ W

(b)(ii) [3] $F = L/(4\pi d^2) = 3.83 \times 10^{26}/(4\pi \times (1.52 \times 10^{11})^2)$
$= 3.83 \times 10^{26}/2.90 \times 10^{23} = 1320$ W m^{-2}

(b)(iii) [1] $\lambda_{max} \propto 1/T$, so $T = 5770/0.581 = 9930$ K

9702/43/M/J/26 Q10 — Sun's luminosity & radiant flux — Wien's law temperature [8 marks]

(a) [2] Wien's displacement law states that the wavelength of maximum emission (of radiation from a star) is inversely proportional to the thermodynamic temperature of its surface: $\lambda_{max} \propto 1/T$.

(b)(i) [2] $L = 4\pi\sigma r^2T^4 = 4\pi \times 5.67\times10^{-8} \times (6.96\times10^8)^2 \times 5770^4$
$L = 4\pi \times 5.67\times10^{-8} \times 4.84\times10^{17} \times 1.11\times10^{15}$
$L = 3.83\times10^{26}$ W

(b)(ii) [3] $F = L/4\pi d^2 = 3.83\times10^{26}/(4\pi \times (1.52\times10^{11})^2)$
$F = 3.83\times10^{26}/2.90\times10^{23} = 1.32\times10^3$ W m^{-2}

(b)(iii) [1] $\lambda_{max} \propto 1/T$, so $T_X = T_S \times \lambda_S/\lambda_X = 5770/0.581$
$T_X = 9.93\times10^3$ K

9702/44/M/J/26 Q10 — Standard candles & a distant star — distance, temperature & radius [7 m

(a)(i) [1] A standard candle is an astronomical object (star) of known luminosity.

(a)(ii) [2] Measure the radiant flux intensity F received from the object, then calculate the distance using the inverse square law: $d = \sqrt{(L/4\pi F)}$, where L is the known luminosity.

(b)(i) [2] The star is moving away from the observer, so its radiation is redshifted — the observed wavelength is greater than the emitted wavelength. The correction removes this Doppler shift, so λ_{star} is the true (emitted) wavelength of maximum intensity.

(b)(ii) [1] Wien's law: $\lambda_{max} \propto 1/T \rightarrow T_{star} = \lambda_0T_0/\lambda_{star}$

(b)(iii) [1] $L = 4\pi\sigma r^2T^4 \rightarrow$ radius $= \sqrt{(L/4\pi\sigma T_{star}^4)}$, where σ is the Stefan–Boltzmann constant.